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AUSTRALIA

CREATE CHANGE

Course Overview

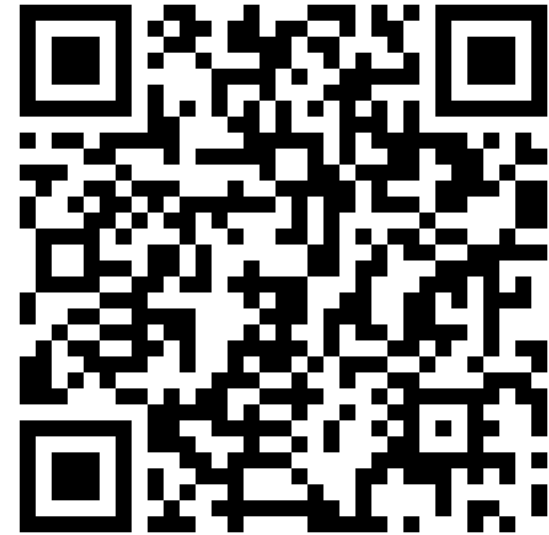
INFS 4205/7205 Advanced Techniques for High Dimensional Data

Yadan Luo

School of EECS, The University of Queensland

Outline

1. Course Background
2. What's **New** This Year
3. Teaching Team
4. Assessments
5. Curse and Blessing of Dimensionality



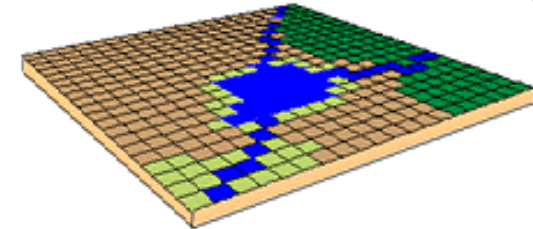
Background Survey 😊

Back to 2007: Spatial and Multimedia Database

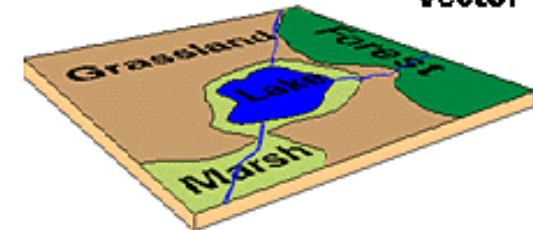
Pre-deep learning era: **data-centric systems**

- Rapid growth of GIS, CAD, Multimedia content
- Databases as the *core* computational bottleneck
- Focus on efficient **storage** and **retrieval** *not* learning
- Strong influence from IR & classical DB systems

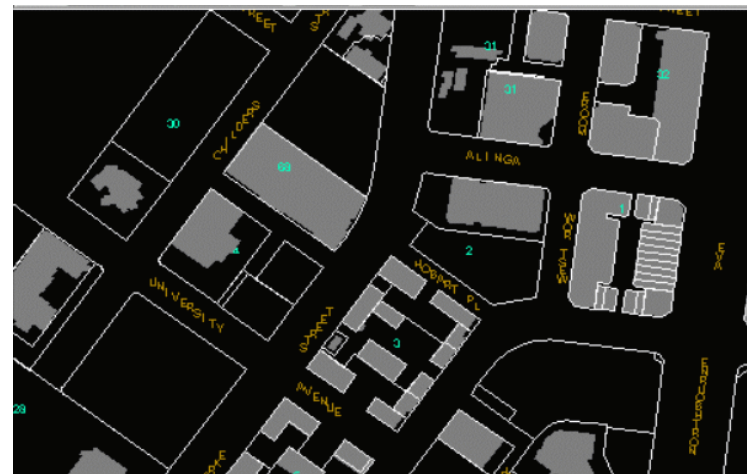
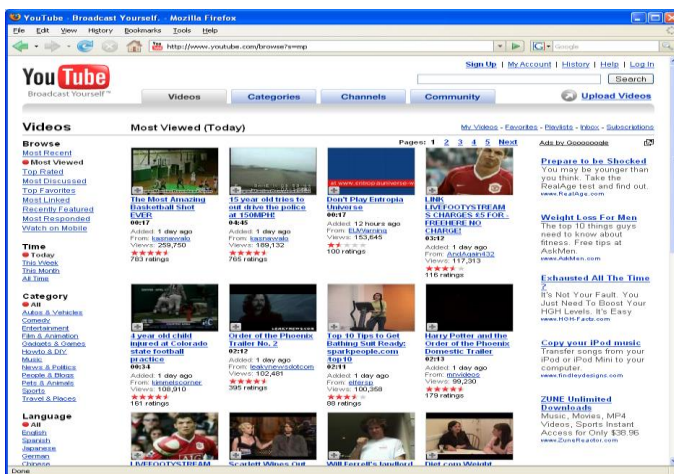
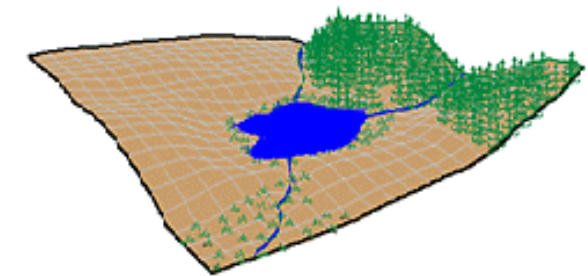
Raster / Image



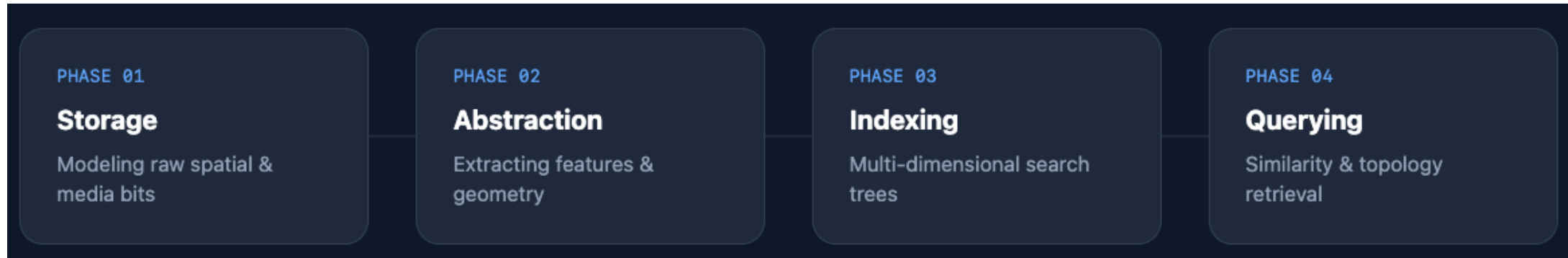
Vector



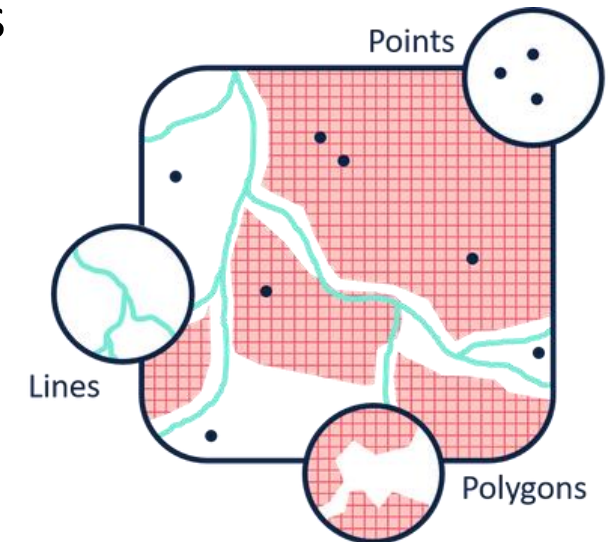
Real World



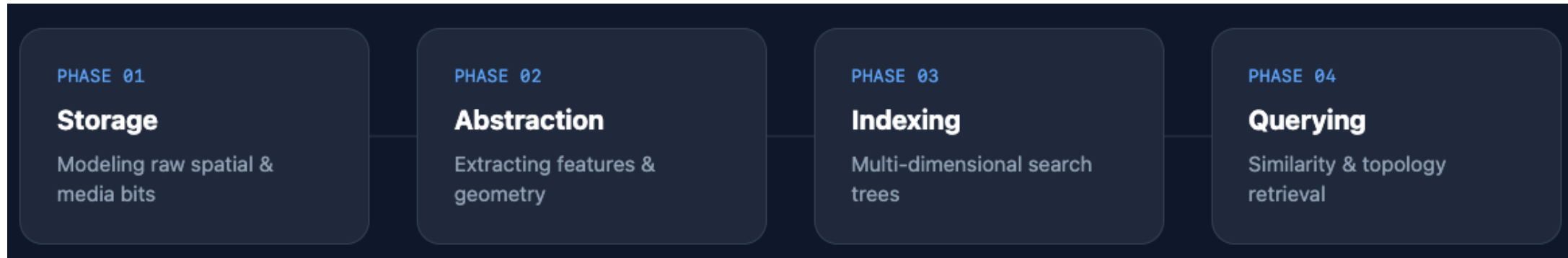
2007: Geometry, Indexing, Querying



1. **Modelling** 2D (coordinates) /3D (trajectory) spatial streams
 - Points, lines, polygons, volumes
2. **Abstraction** (feature extraction before querying)



2007: Geometry, Indexing, Querying



1. Modelling 2D (coordinates) /3D (trajectory) spatial streams

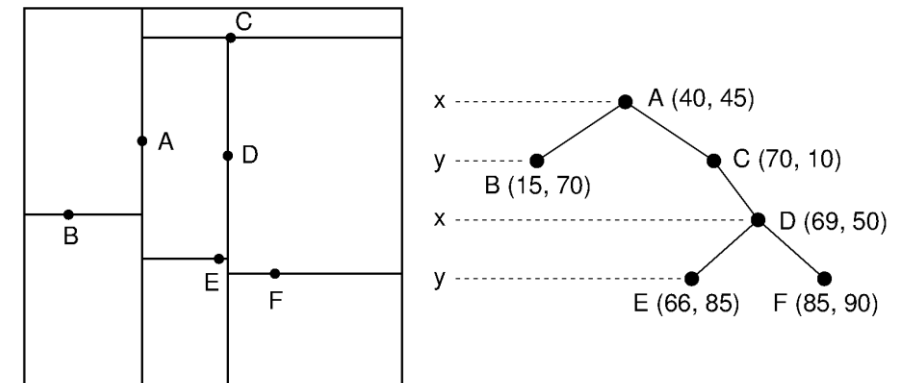
- Points, lines, polygons, volumes

2. Abstraction (feature extraction before querying)

3. Spatial Indexing (R-tree, KD-tree, Quad-tree)

4. Query Processing (range query, kNN, skyline)

- Design goal: **minimize disk I/O**



2017-2025: AI enters Databases

Traditional DBMS struggle with High-dim Data... as **Curse of Dimensionality** breaks classical indexing

1. How to handle HD data with AI?

- Dimension reduction, feature selection (project **high**-dim to **low**-dim manifold)
- Preserve structure: distance / topology / neighborhood



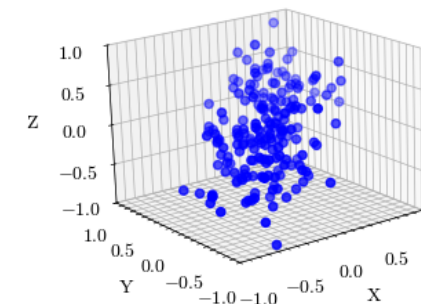
☐ include secondary database models

8 systems in ranking, February 2024

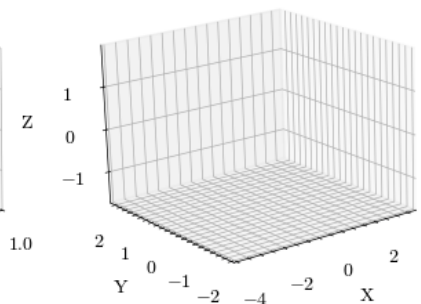
Rank			DBMS	Database Model	Score		
Feb 2024	Jan 2024	Feb 2023			Feb 2024	Jan 2024	Feb 2023
1.	1.	1.	PostGIS	Spatial, Multi-model 	25.88	-0.78	-3.93
2.	2.	2.	Aerospike 	Multi-model 	7.09	+0.33	+0.53
3.	3.	3.	Spatialite	Spatial, Multi-model 	2.24	-0.06	-0.01
4.	4.	4.	GeoMesa	Spatial	0.92	-0.09	-0.02
5.	5.	5.	CrateDB 	Multi-model 	0.91	-0.09	0.00
6.	6.		Apache Sedona	Spatial	0.29	-0.12	
7.	7.	↓ 6.	H2GIS	Spatial, Multi-model 	0.03	-0.12	-0.17
8.	8.	↓ 7.	SpaceTime	Spatial, Multi-model 	0.00	±0.00	±0.00

Principal Component Analysis

PC Hyperplanes



Projected Data

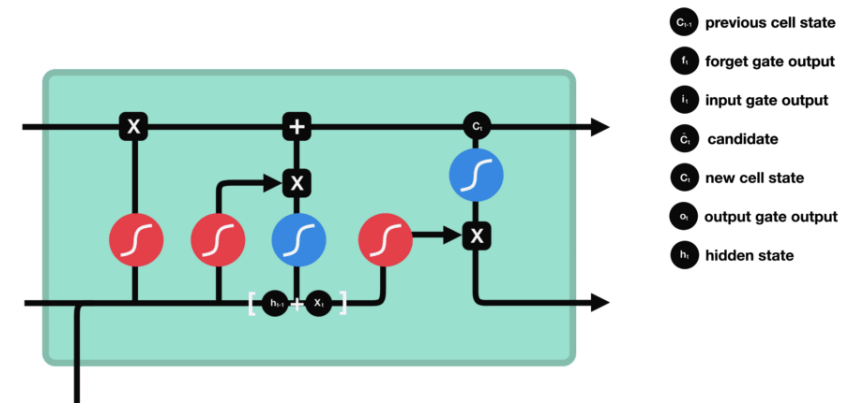
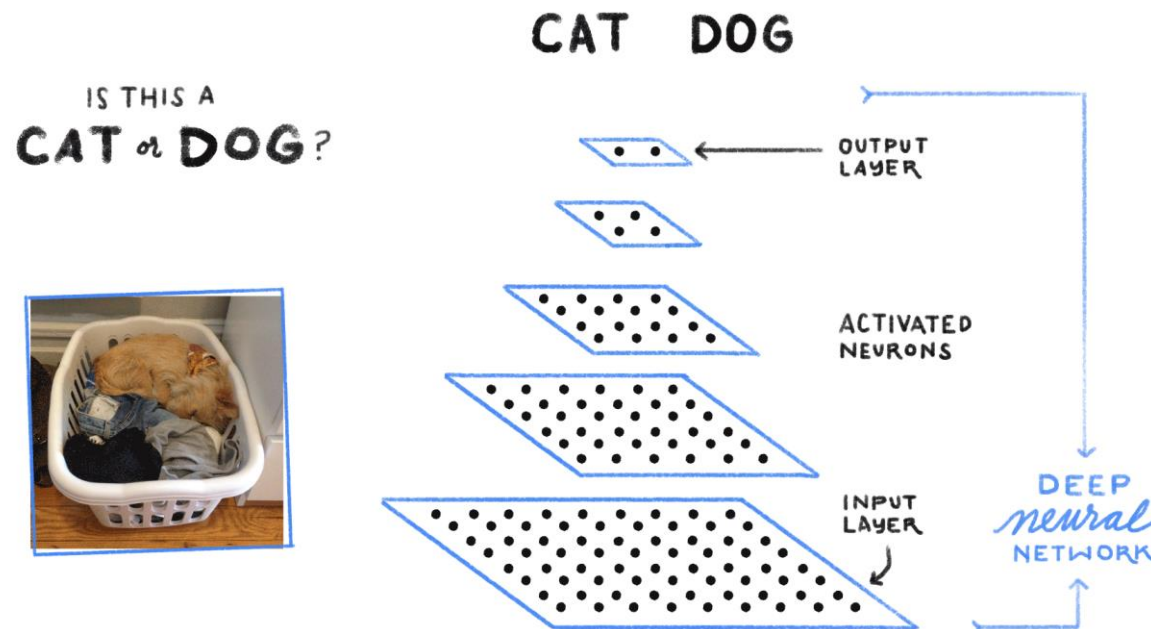


2017-2025: AI enters Databases

Traditional DBMS struggle with High-dim Data... as **Curse of Dimensionality** breaks classical indexing

1. How to handle HD data with AI?

- Deep neural networks are introduced to help *replace* hand-crafted *feature engineering*



LSTM / GNN to model mobility patterns

2017-2025: AI enters Databases

2. What unique challenges does AI bring to DBMS?

From structured data to **embeddings...**

- Rows & columns → dense vectors
- Schema-less representations
- Semantics encoded implicitly

Change! Change! Change!

Data is continuously updated (embeddings drift)

Models change → vectors change

Queries are interactive & multi-stage

👉 *Index maintenance becomes a first-class problem*

 Pinecone		Proprietary composite index
 milvus /  zilliz		Flat, Annoy, IVF, HNSW/RHNSW (Flat/PQ), DiskANN
 Weaviate		Customized HNSW, HNSW (PQ), DiskANN (in progress...)
 drant		Customized HNSW
 chroma		HNSW
 LanceDB		IVF (PQ), DiskANN (in progress...)
 vespa		HNSW + BM25 hybrid
 Vald		NGT
 elasticsearch		Flat (brute force), HNSW
 redis		Flat (brute force), HNSW
 pgvector		IVF (Flat), IVF (PQ) in progress...

By way of the Background...

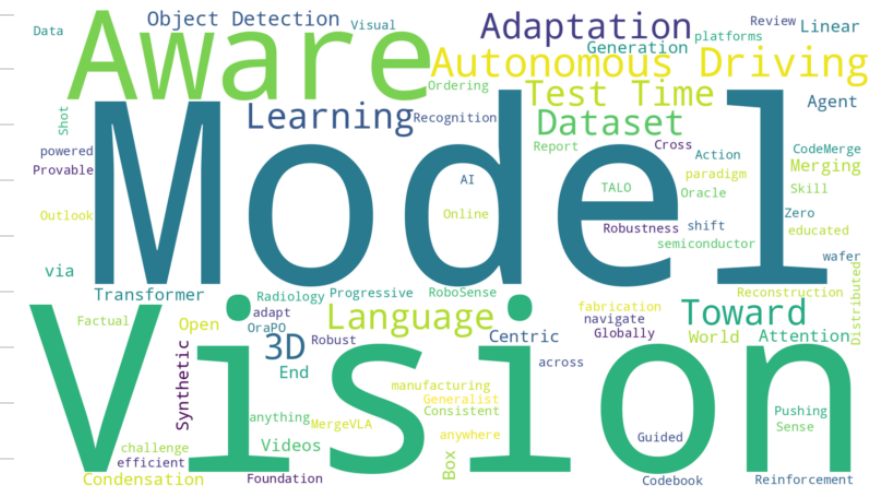
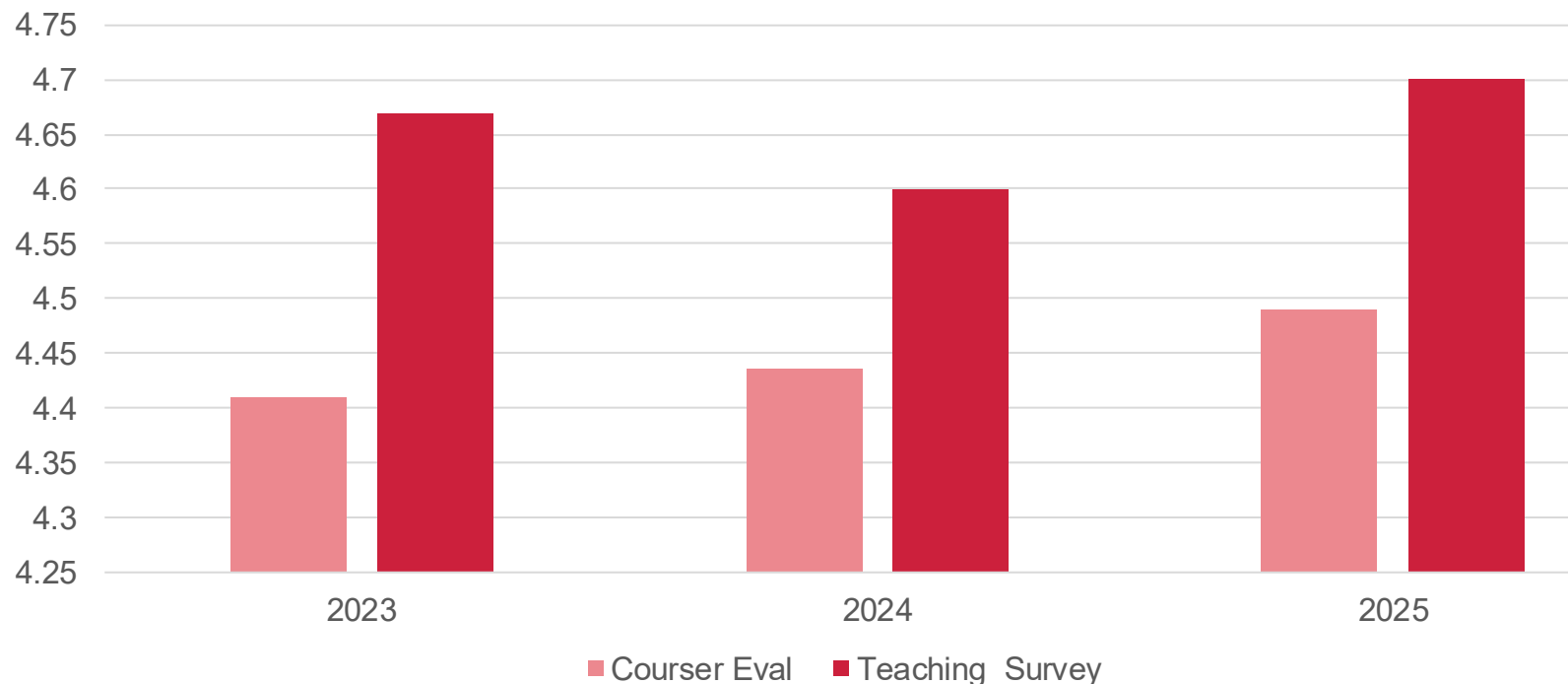
- I took over this course since 2023;
- Valuable feedback motivated the progressing of this course



Senior Lecturer / INTJ
Computer Science, School of EECS

<https://luoyadan.github.io/>

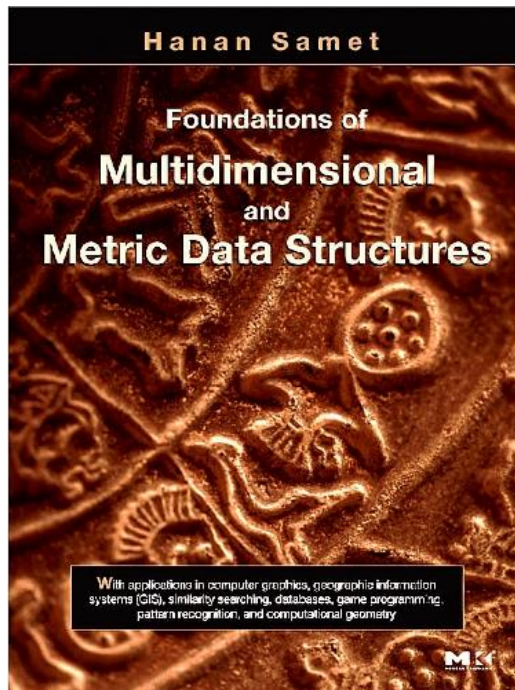
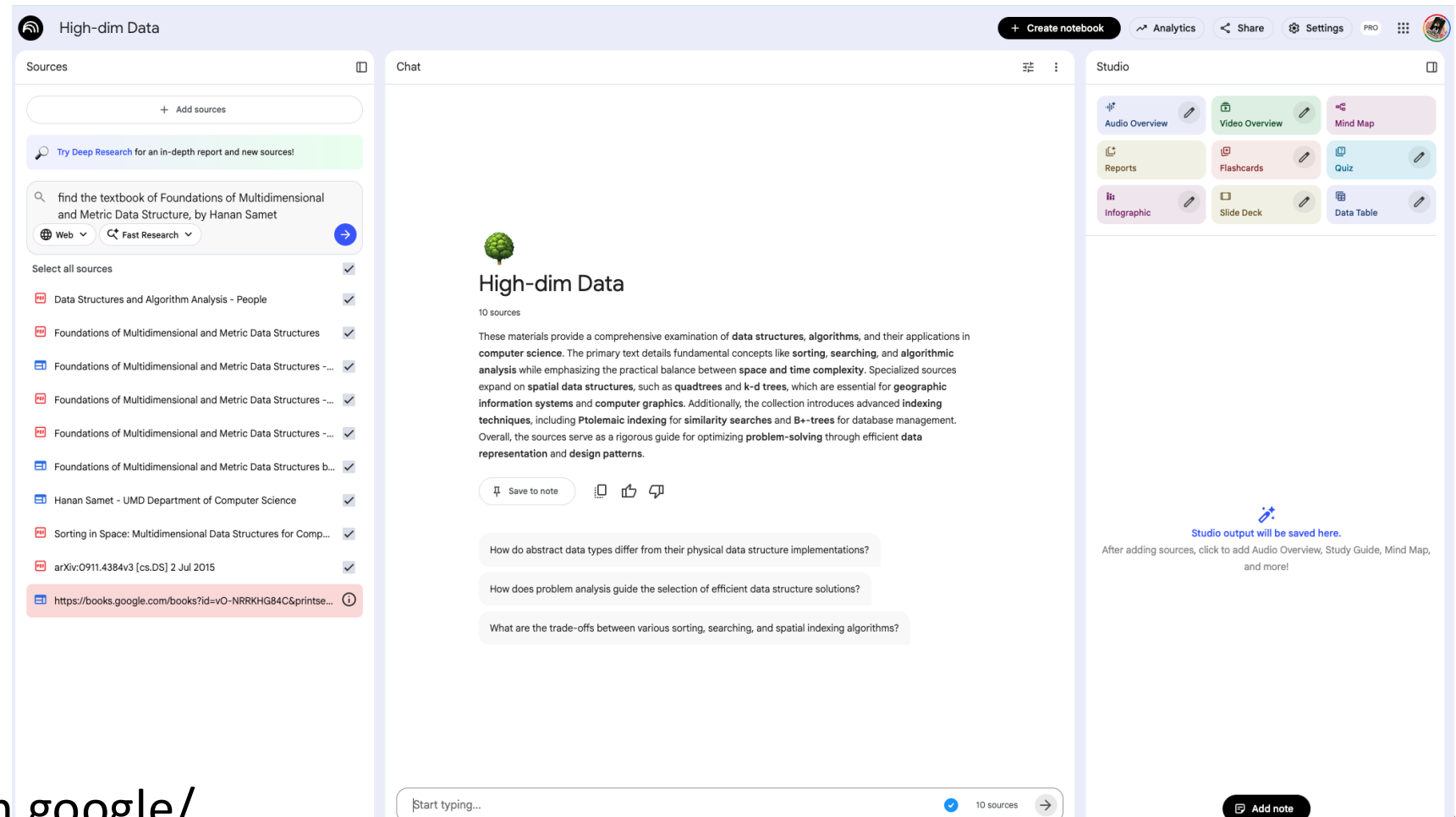
SeCAT Evaluation of INFS4205/7205 (2023-2025)



2026: What should we learn from this course?

- AI has transformed our way of teaching and learning (NotebookLM)

 NotebookLM

The screenshot shows the NotebookLM interface with a notebook titled "High-dim Data". The "Sources" panel on the left lists 10 sources, including "Data Structures and Algorithm Analysis - People", "Foundations of Multidimensional and Metric Data Structures", and "Hanan Samet - UMD Department of Computer Science". The "Chat" panel in the center displays a summary of the notebook's content, mentioning "data structures, algorithms, and their applications in computer science". The "Studio" panel on the right offers various output options: Audio Overview, Video Overview, Mind Map, Reports, Flashcards, Quiz, Infographic, Slide Deck, and Data Table. A message at the bottom of the Studio panel states: "Studio output will be saved here. After adding sources, click to add Audio Overview, Study Guide, Mind Map, and more!"

<https://notebooklm.google/>

2026: What should we learn from this course?

Chat

What are the trade-offs between various sorting, searching, and spatial indexing algorithms?

Today • 5:39 PM

how to build kd tree

To build a k-d tree (k-dimensional tree), you construct a binary tree that extends the concept of a standard Binary Search Tree (BST) to handle multidimensional data, such as xy -coordinates ¹. The construction process relies on **object space decomposition**, meaning the tree's structure and split points are determined by the specific data values of the objects stored and the order in which they are inserted ², ³.

The fundamental logic for building the tree revolves on the use of a **discriminator** at each level:

- 1. Define the Discriminator:** unlike a BST which always compares the same single key, a k-d tree changes the comparison key at every level of depth. For k dimensions, the discriminator at level i is defined as $i \bmod k$ ¹.
 - For a 2-dimensional set (using x and y coordinates):
 - Level 0 (Root):** Splits based on the x -coordinate ($0 \bmod 2 = 0$).
 - Level 1:** Splits based on the y -coordinate ($1 \bmod 2 = 1$).
 - Level 2:** Splits based on the x -coordinate, and so on ¹.
- 2. Insertion Process:** To build the tree, you insert records one by one using a recursive search procedure:
 - Compare the new record's value for the current level's discriminator against the value stored in the current node ⁴.
 - If the new record's value is less than the node's value, proceed to the left subtree; otherwise, proceed to the right subtree ¹.
 - As you move down a level, switch to the next discriminator in the sequence (e.g., from x to y) ⁵.
 - When you reach a **NULL** pointer, attach the new node at that location ⁶.

Example of Construction: Consider inserting a record P at location $(10, 50)$ into a tree where the root node B is located at $(15, 70)$:

Studio > App

Algorithms Quiz

Based on 10 sources

6 / 12

What is a primary distinction between the R-tree and the R+-tree spatial data structures?

A. R+-trees have a fixed maximum depth, unlike R-trees.

B. R+-trees can only be used for point data, whereas R-trees can handle regions.

C. R-trees allow overlapping bounding boxes for sibling nodes, while R+-trees enforce disjoint (non-overlapping) rectangular regions.
✓ **That's right!**
This lack of overlap in R+-trees prevents a spatial object from being contained in multiple branches, which can simplify some search operations at the cost of more complex insertion and deletion.

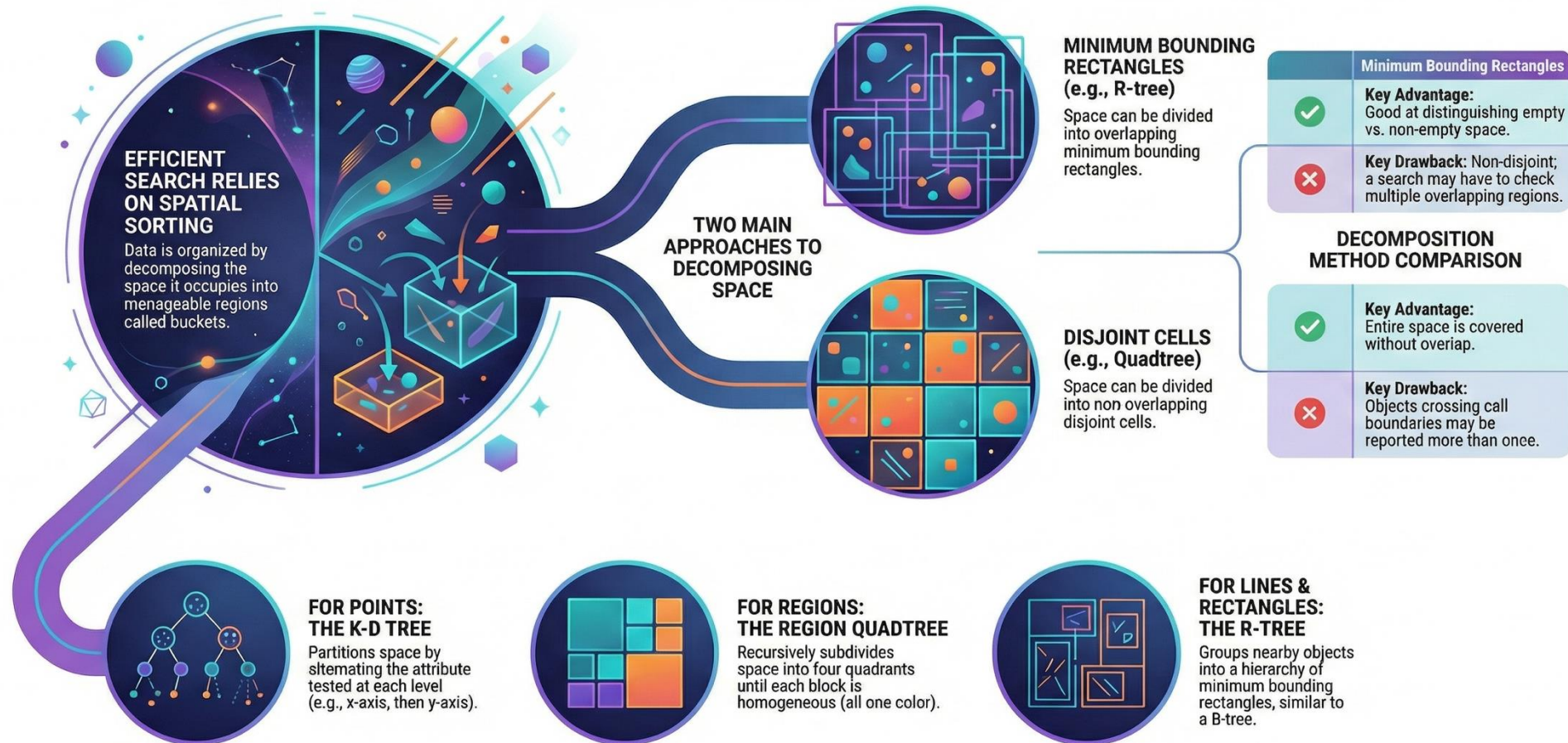
D. R-trees are based on a uniform grid decomposition of space.

[🔗 Explain](#)
[Previous](#)
[Next](#)

2026: What should we learn from this course?

SORTING IN SPACE: A GUIDE TO SPATIAL DATA STRUCTURES

Effectively querying spatial data—like points, lines, and regions—requires specialized data structures. Unlike simple lists, these structures "sort" data by its location, decomposing space hierarchically to enable rapid operations like finding the nearest neighbor or objects within a specific area.



2026: What should we learn from this course?

Sorting Geometry From Lists to Networks

Based on 10 sources

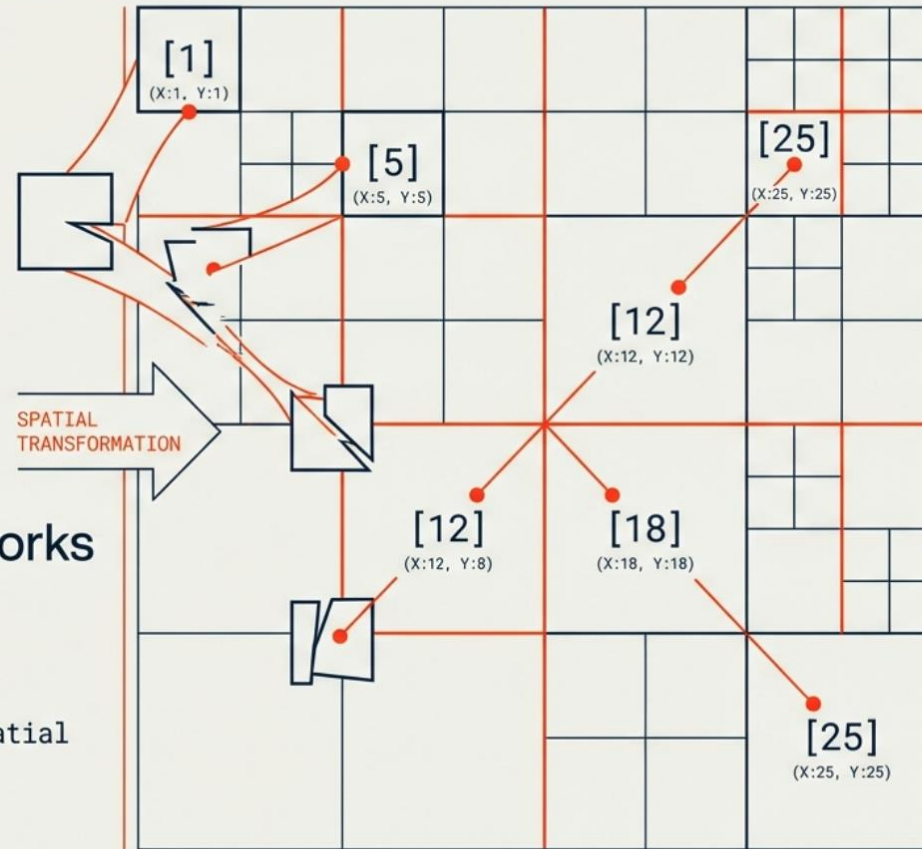


Sorting in Space

From Linear Lists to
Multidimensional Networks

A Deep Dive into Hierarchical Spatial
Data Structures and Algorithms

Based on the work of Hanan Samet and
Clifford Shaffer



2026: What's new?

-  “In the age of AI, what skills matter most?”

2026: What's new?

- 🙄 “In the age of AI, what skills matter most?”

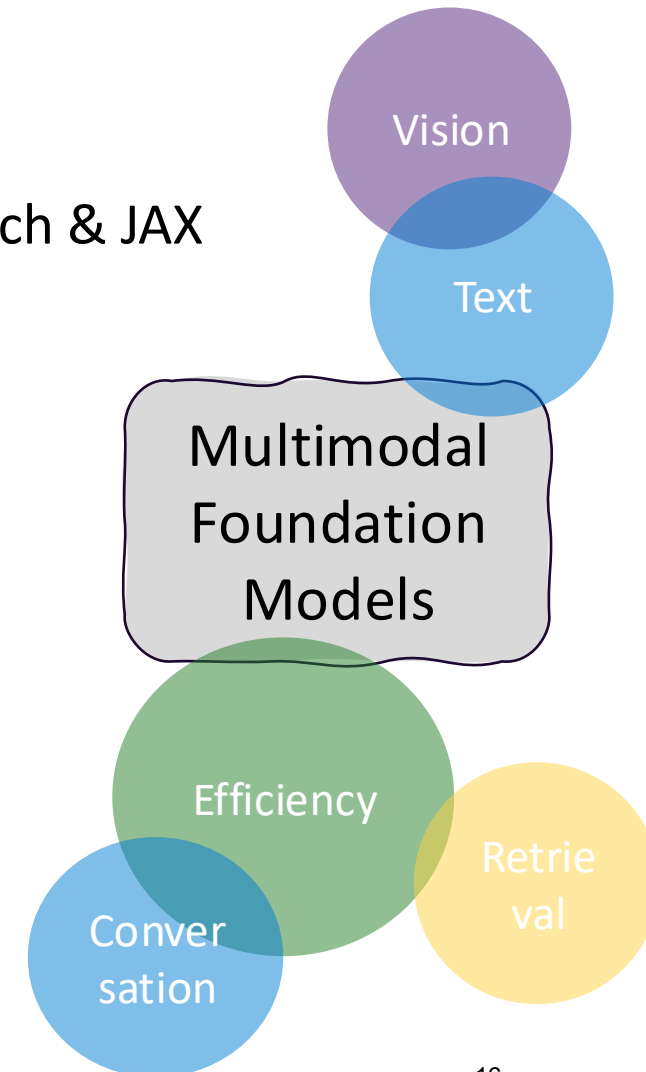
1 Understand the tool before you use it

- **knowing** the underlying mechanism of AI (e.g., LLMs, VLMs) through *the lens of high-dim data*
- **Familiar** with the terminology and progress (**in a systematic way!**)
 - Transformers, tokens, batch, softmax, scaling, alignment, RAG, agents, vectorDB etc.
 - Then think: What leads to future research in AI? Any high-dim challenges persist?
 - Any opportunities? (tools change fast, mechanisms last)

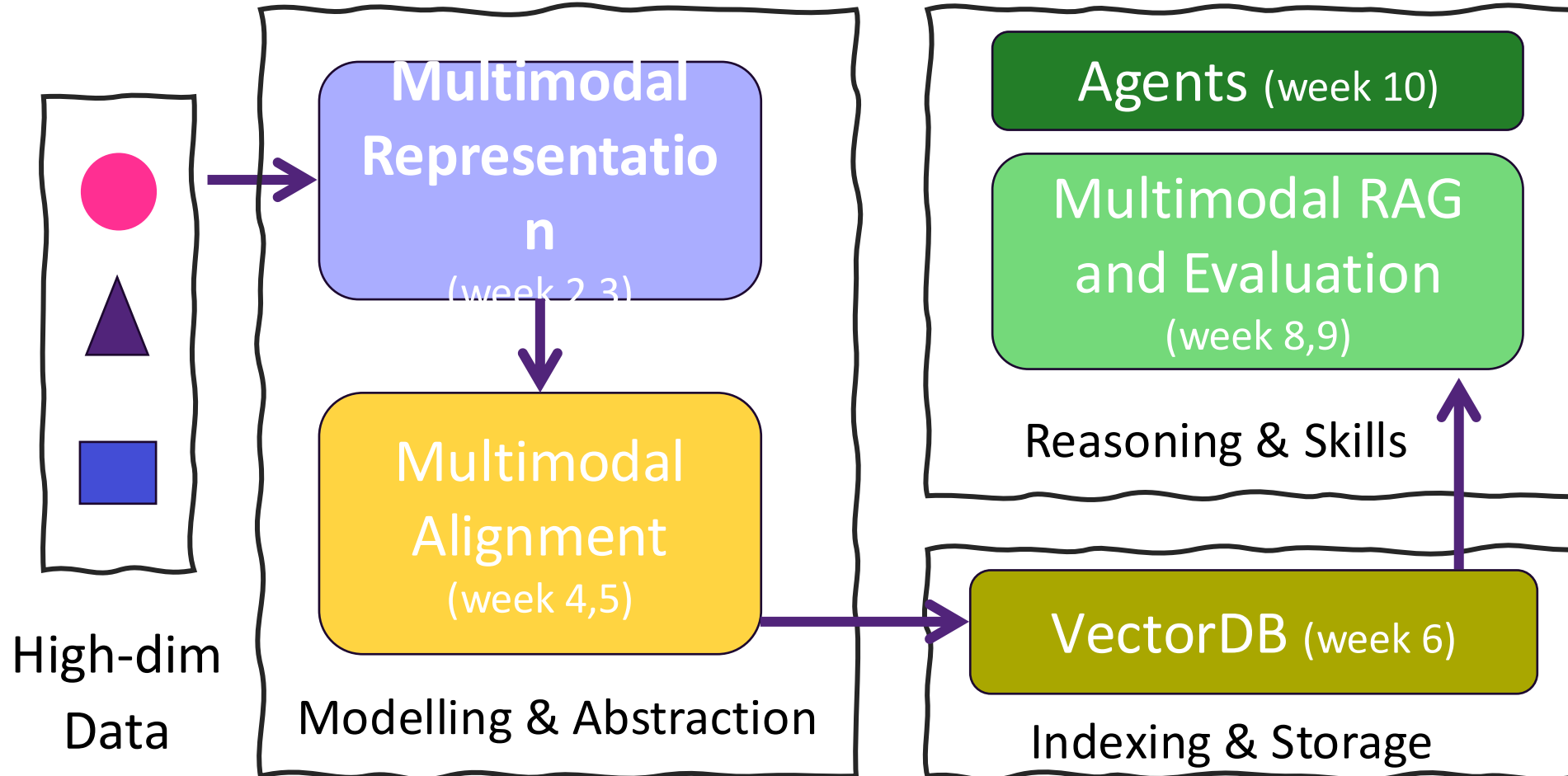
2026: What's new?

Course Design: **Different** from courses including

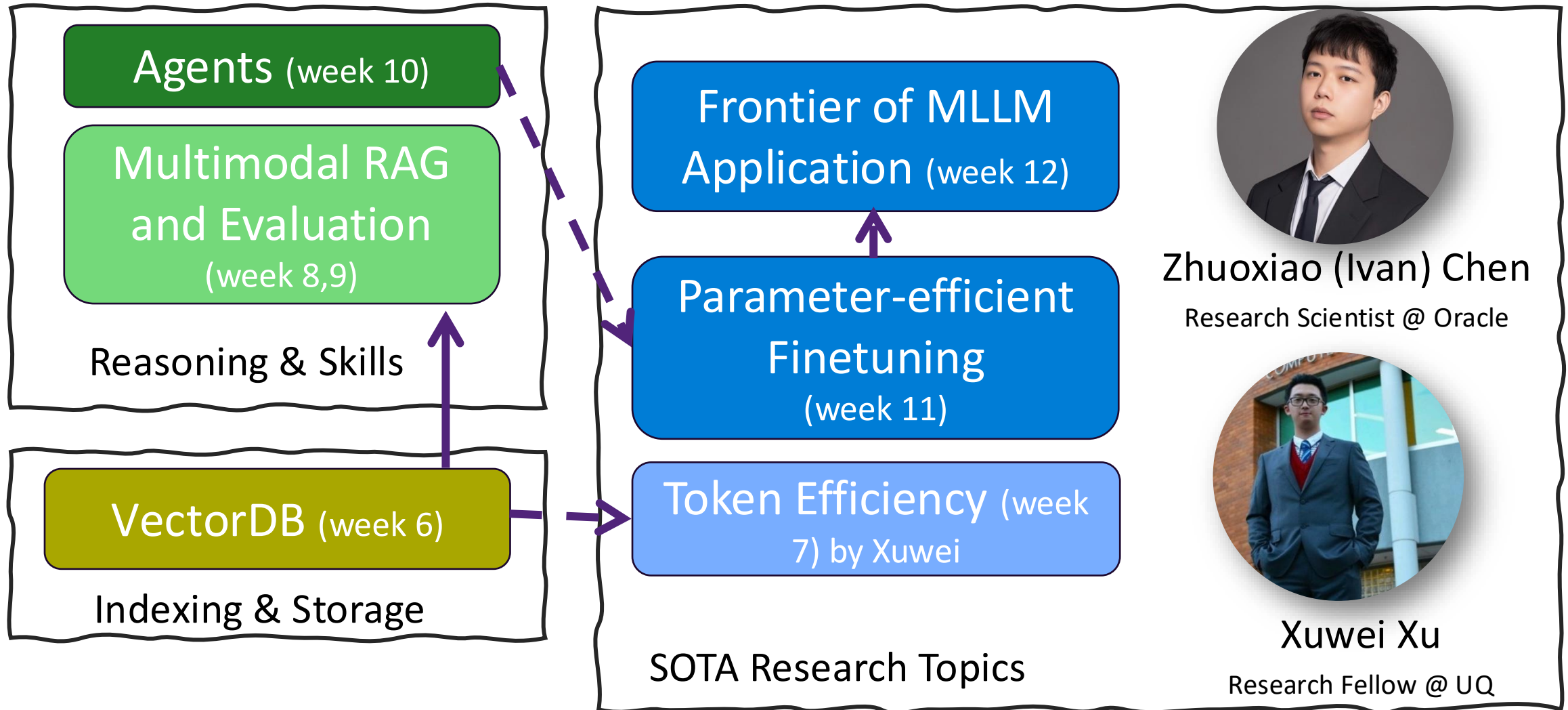
- **(COMP3710) Pattern Recognition and Analysis**
 - Classification (random forests), deep learning (image analysis), PyTorch & JAX
- **(COMP3702) Artificial Intelligence**
 - Search, Planning, RL
- **(INFS7410) Information Retrieval and Web Search**
 - BM25, Retrieval systems
- **(COMP4703) Natural Language Processing**
 - Machine translation, sentiment analysis, question answering
- **(INFS4203) Data Mining and (COMP4702) Machine Learning**
 - Clustering, Anomaly detection, light touch on deep learning
 - Dimension reduction (PCA etc.)



Tentative Syllabus (updated in BB)



Tentative Syllabus (updated in BB)



Tentative Syllabus (updated in BB)

Week #	Lec Date	Lecture	Speaker	Notes
Week 1	24 Feb	Course Overview	Yadan	What is Dimension and Modality,
Week 2	3 Mar	Multimodal Representation (Part I.)	Yadan	Early Models for feature encoding txt; Transformers &-> then for visual ViT
Week 3	10 Mar	Multimodal Representation (Part II.)	Yadan	Curse of Dimensionality; -> scaling -> early self-supervised representation learning (DiNO)
Week 4	17 Mar	Multimodal Alignment	Yadan	Contrastive learning, CLIP, BLIP, BLIP-2 (Q-Former)
Week 5	24 Mar	From Alignment to Understanding: MLLM	Yadan	Projection Layer, Visual Instruction Tuning (LLaVA, Flamingo, Qwen-VL)
Week 6	31 Mar	High-dimensional Data Indexing + Midterm Exam	Yadan	Vector DB; ANN; HNSW (faiss)
Mid Break	7 Apr			
Week 7	14 Apr	High-dimensional Data Efficiency	Xuwei	Information Bottleneck; Token reduction, Linear Attention
Week 8	21 Apr	Multimodal RAG	Yadan	CoT; ColBERT, ColPali
Week 9	28 Apr	Evaluation of RAG	Yadan	Hallucination Detection (OOD); RAGAS, TruLens, G-Eval
Week 10	5 May	Agents	Yadan	AutoGPT, LangGraph, multi-agent systems (LangGraph)
Week 11	12 May	PEFT & Personalisation	Ivan	LoRA, DoRA, Adapters, Model Merging
Week 12	19 May	Frontier of MLLM Applications	Ivan	Medical Report Generation
Week 13	26 May	Oral Exam	All	

Teaching Team

Tutors



Danny Wang
(Tutor Head)



Yan Jiang



Zhizhen Zhang



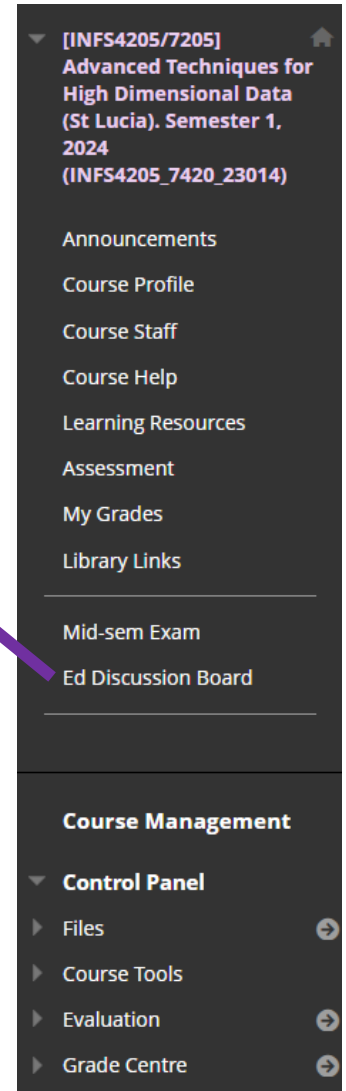
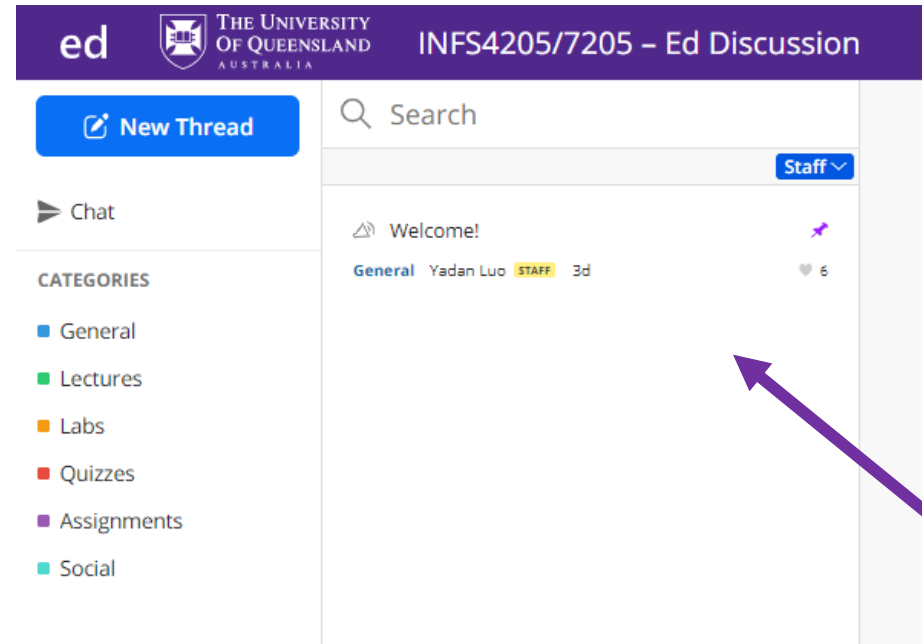
Fengyi Zhang



Xiangyu Sun

Seek Assistance?

- Post your questions on ED
- Email us: y.luo@uq.edu.au
- Tutorial will cover practical interview questions
 - Show Chain of Thought 😊



**Attendance is not mandatory for either lectures or tutorials.
Nevertheless, you are highly encouraged to do so.**

2026: What's new?

- 🙄 “In the age of AI, what skills matter most?”

2 Ask the right questions

- Moving from **textbook-driven** to **question-driven** thinking
- No requisite for this course, but also no excuses 😊
 - learn what you need, *when you need it*
- AI progress is often about asking better questions, not training bigger models
- Capacity of designing “**good prompts**”?
- From “passive learning” to “active learning”

2026: What's new?



Assessment Items I. AI-Assisted Exploration

- **AI-Assisted Exploration** (5 submissions * 5% = 25%)
 - **Submission:** Individual, via Blackboard
 - **Use of AI:** not limited to ChatGPT, Gemini ... (in English)
 - **X** AI output itself is not assessed
 - **✓** Marks are based on your thinking and reflection

Instead of memorising techniques, you will:

- ❖ identify what you *do not fully understand*,
- ❖ ask meaningful questions,
- ❖ use AI as a learning aid,
- ❖ and reflect critically on what you learned.



Assessment Items I. Reflective Inquiry (AI-Assisted)

- **What to submit?** -> must include 3 short sections (at most 1 page)

1. Your question

- One clear, non-trivial question related to the lecture topic
- Focus on why, limitations, comparisons, or failure cases

2. AI-assisted Exploration (optional: one screenshot)

- Briefly summarise how AI helped you explore the question
- What was helpful? What was unclear or incomplete?

3. Reflection

- What did the AI miss or oversimplify?
- How did lectures or other readings refine your understanding?
- What did you learn in the end

Assessment Items I. Reflective Inquiry (AI-Assisted)

• Rubric

Grade	Descriptor
 Genius	Insightful question, purposeful AI use, and clear critical reflection showing learning beyond AI and lecture slides <i>(you learned something I did not cover, or you caught AI's mistakes!)</i>
 Excellent	Meaningful question and appropriate AI use with some critical reflection
 Very Good	Relevant but surface-level question with limited reflection or some <i>over-reliance</i> on AI
 Needs Work	Trivial or weakly connected question with minimal reflection
 Come on!	Vague, copied, or missing question; little evidence of independent thinking

Assessment Items I. Reflective Inquiry (AI-Assisted)

Template:

- A good question comes from something concrete
- Don't just ask **What**; ask **Why, How, What if**

"When learning about ____, I noticed ____."

Weak

Strong

What is tokenization?

Why does tokenization affect model bias?

What is RAG?

How does retrieval quality limit RAG reasoning?

What is overfitting?

Why can large models still generalize?

"AI explained ____, which was not covered in class. How does this relate to ____?"

What is the trade-off between model size and retrieval quality in RAG systems?

If AI hallucination cannot be fully eliminated, how should systems be designed safely?

Assessment Items I. Reflective Inquiry (AI-Assisted)

★ Genius

The lecture presents attention as solving long-range dependencies, but AI explanations often attribute this to positional encoding. Which component is truly responsible, and how do ablation studies disentangle their roles?

👍 Excellent

AI explained that RAG reduces hallucination, but some papers report persistent errors. Why does retrieval not fully prevent hallucination?

😊 Very Good

How does RAG work in large language models?

⚠️ Needs Work

What is attention?

✗ Come on

Explain lecture 3.



Assessment Items II. In-Semester Quiz

- **In-Semester Quiz (Week 6)** (10%)
 - **Submission:** In-class quiz, 60min, paper-based
 - **Closed-book:** Cover Week 2-Week 5

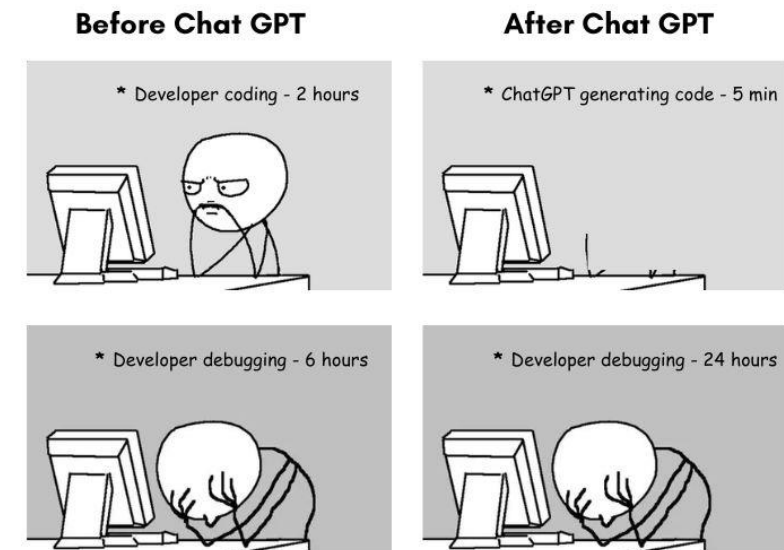
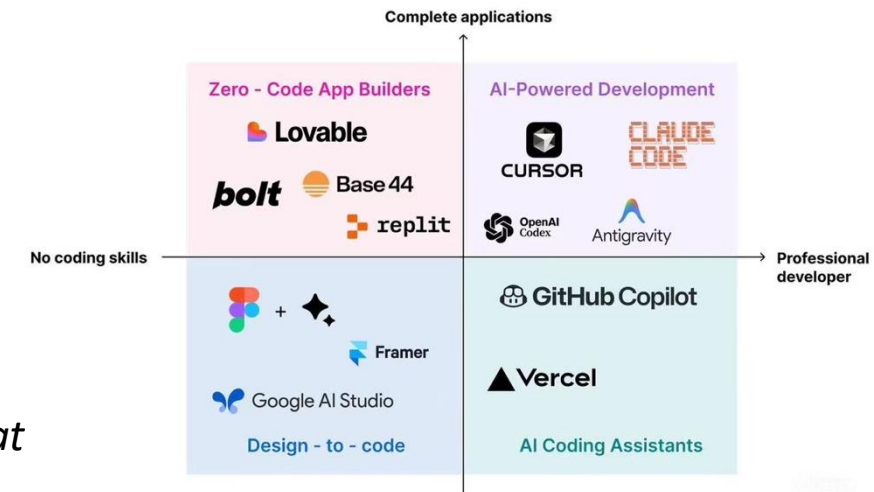
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2026: What's new?

- 🤔 “In the age of AI, what skills matter most?”

3 Vibe Coding

- Starting from **intent**, not syntax (→ *What problem am I solving? What should the system do?*)
- Using AI to prototype fast, not to think for you (→ *You **design** the logic!*)
- **What you will practice** from this course:
 - Guide AI to do works with plans (*break tasks into steps before asking for code*)
 - Unit testing as a habit! (→ small tests, frequent checks, early failure)
 - Debug with understanding (not just ctrl c+v 😊)



Assessment Items III - Project

Project (35%): Personalised Multimodal Chatbot

Why Vibe Coding?



[Officially Live] Meta's New AI-enabled Coding Round: What I've Learned So Far

Meta just rolled out a new AI-enabled coding round that replaces one of the traditional onsite coding rounds (two LeetCode-style problems in 35 mins). Instead of classic meta-tagged algo problems and their variants, from what I've gathered you'll get one of three scenarios: building a feature from scratch, extending an unfamiliar multi-file codebase, or debugging broken code under time pressure. All with AI assist, plus real execution and testing (Python candidates: brush up on `unittest` if you haven't already). I started seeing members of my interview prep Discord getting this round over the past week or 2, and since there's not much info out there yet, I spoke to them to gather as much insight as possible. Below I discuss what I've gathered so far, you can find practice questions [here](#), hope it helps.

The Basics

- **60-minute CodePad session** with an AI-assist chat window and model switcher (similar to Copilot Chat, but simpler).
 - **Confirmed available in interviews:** GPT-4o mini, Claude 3.5 Haiku, Llama 4 Maverick
 - **Not confirmed in interviews (but supported by CodePad):** GPT-5, Gemini 2.5 Pro, Claude 4.5 Haiku, Claude 4 Sonnet, Claude 4.5 Sonnet
So you might have access to these too. It depends on what Meta enables in your interview environment.
- **One thematic question** with multiple checkpoints or stages (so it can be a multi-part question), not two separate LeetCode problems. Empirically clearing 3 of the checkpoints seems to be necessary (not sufficient) to clear this round. I've seen people that cleared 3 checkpoints get called to redo though.
- You get a **mini multi-file codebase** (for Python: multiple `.py` files plus `requirements.txt`)
- **You can run and debug code** in real-time.
- Started appearing **early October 2025** for SWE and ML; likely rolling out to Production Engineers soon

...

What This Round Actually Tests

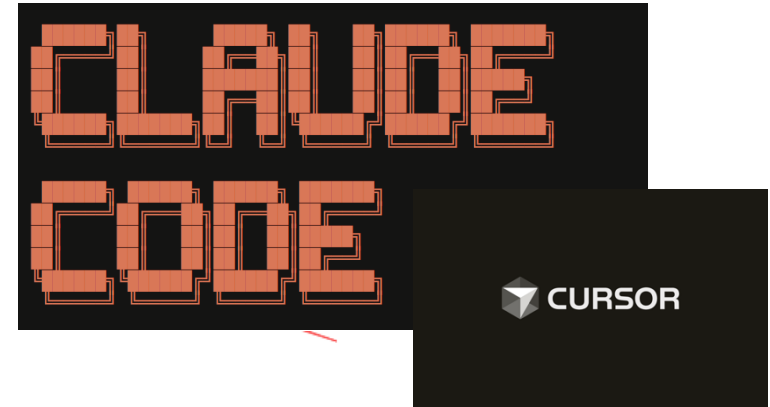
What do we look for?

The AI-Enabled Coding Interview will assess your performance on the following four focus areas: Problem Solving, Code Development and Understanding, Verification and Debugging, and Technical Communication.

- **Problem Solving:** Are you able to clarify and refine problem statements? Can you generate solutions to open-ended and quantitative problems?
- **Code Development and Understanding:** Are you able to navigate a codebase to develop and build on working code structures and to evaluate the quality of produced code? Can you analyze and improve code quality and maintainability? Does code work as intended after it is executed?
- **Verification and Debugging:** Can you find and mitigate errors to ensure code runs/functions as intended? Are you able to verify solutions meet specified requirements, leveraging test/edge cases and handling errors and exceptions? How well do your unit tests run?
- **Technical Communication:** How well can you communicate reasoning, discuss technical ideas, ask thoughtful questions, and incorporate feedback?

What You Need to Know About the AI

- **It is not a frontier reasoning model.** Expect hallucinations, suboptimal suggestions, and missed edge cases
- The AI can see all code in your editor (no copy-paste needed)
- **This is not a prompting test.** You are evaluated on problem-solving and verification, not AI expertise
- Some candidates barely use it and excel; others use it heavily for boilerplate and also excel
- The AI is great for: boilerplate, parsing, scaffolding, heavy typing, and help with debugging and navigating the codebase.
- AI struggles with: 100% accuracy, algorithmic optimality, edge cases, and deep reasoning



Assessment Items III - Project

Project (35%): Personalised Multimodal Chatbot



What you are building:

- A **multimodal chatbot** backed by your personal knowledge base
- The chatbot retrieves relevant content and answers questions using **multimodal RAG**
- The focus is on **indexing, querying, and evaluation**, not UI polish



challenging problems with real-world impact.

Qualifications and Experience

- Demonstrated experience in designing and developing innovative AI models, algorithms and frameworks, with a strong publication track record in top conferences and journals.
- Deep technical knowledge of Machine Learning and Deep Learning architectures, such as Transformers, training methodologies, and optimization techniques.
- Strong familiarity with the latest advancements in Large Language Models (LLM) and generative AI, including but not limited to agent technology, memory, multimodality, reinforcement fine-tuning, pre-training, fine-tuning, alignment, reasoning, RAG, automated prompt/workflow optimisation, advanced prompt engineering, and their applications such as dialogue systems, information extraction, question answering, language generation/summarization.
- Practical experience with emerging LLM frameworks and plugins such as LangChain, LlamaIndex, VectorStores, Retrievers, LLM Cache, and LMQL, showcasing your ability to stay at the forefront of AI technology.
- Knowledge of the healthcare domain and experience in delivering AI models specifically for healthcare applications are highly desirable.

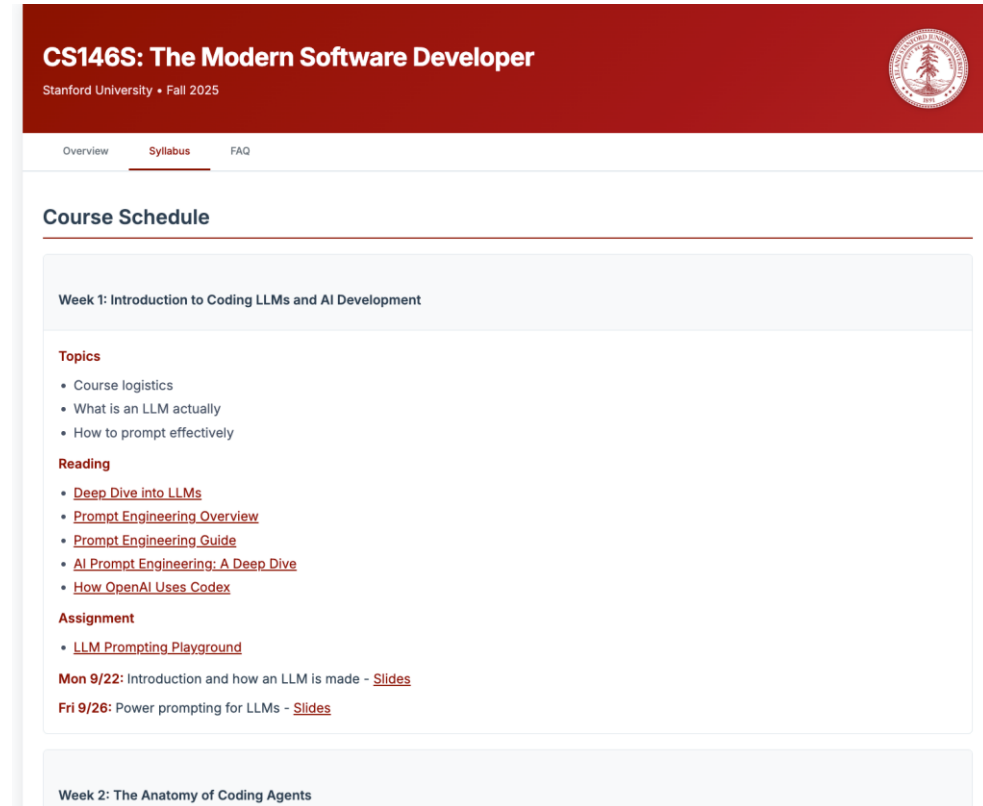
Education

Assessment Items - Project

Project (35%): Personalised Multimodal Chatbot

Learning Resources for Vibe Coding

- Stanford CS146S: The Modern Software Developer
 - More advanced tips on MCP, coding agent
 - Automated UI and App Building
- Build your first AI-demo -> promote on LinkedIn/CV



The screenshot shows the course page for CS146S: The Modern Software Developer at Stanford University, Fall 2025. The page has a red header with the course title and university logo. Below the header, there are tabs for Overview, Syllabus, and FAQ. The main content area is titled 'Course Schedule' and lists the topics for Week 1: Introduction to Coding LLMs and AI Development. The topics include Course logistics, What is an LLM actually, and How to prompt effectively. The reading list includes Deep Dive into LLMs, Prompt Engineering Overview, Prompt Engineering Guide, AI Prompt Engineering: A Deep Dive, and How OpenAI Uses Codex. The assignment is LLM Prompting Playground. The dates for the first two sessions are Mon 9/22 and Fri 9/26, with links to slides for each.

CS146S: The Modern Software Developer
Stanford University • Fall 2025

Overview Syllabus FAQ

Course Schedule

Week 1: Introduction to Coding LLMs and AI Development

Topics

- Course logistics
- What is an LLM actually
- How to prompt effectively

Reading

- [Deep Dive into LLMs](#)
- [Prompt Engineering Overview](#)
- [Prompt Engineering Guide](#)
- [AI Prompt Engineering: A Deep Dive](#)
- [How OpenAI Uses Codex](#)

Assignment

- [LLM Prompting Playground](#)

Mon 9/22: Introduction and how an LLM is made - [Slides](#)

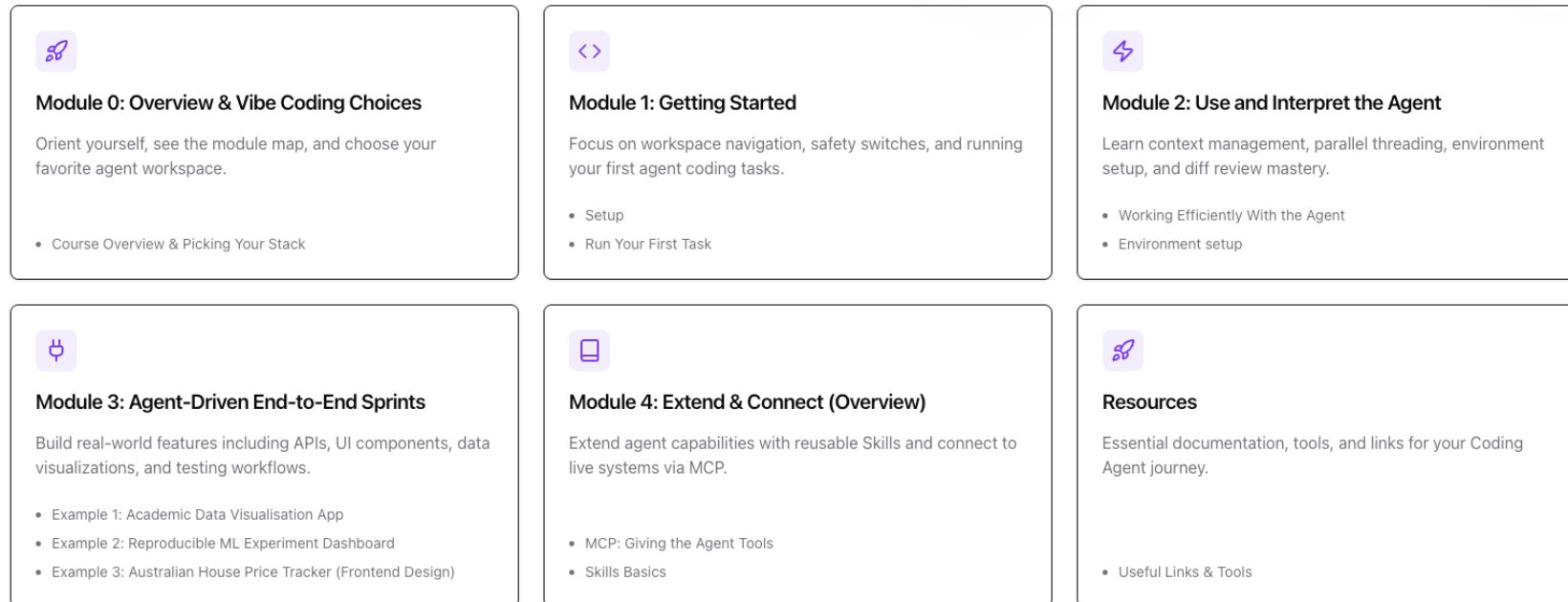
Fri 9/26: Power prompting for LLMs - [Slides](#)

Week 2: The Anatomy of Coding Agents

Assessment Items - Project

Project (35%): Personalised Multimodal Chatbot

- 3 modular “101-level” Intro to use cursor (cred. Danny Wang)



Team: **Danny Wang***, Yadan Luo*, Zhuoxiao Chen, Yan Jiang, Xiangyu Sun, Xuwei Xu, Fengyi Zhang, Zhizhen Zhang.

* Project Credit, * Coordinator.

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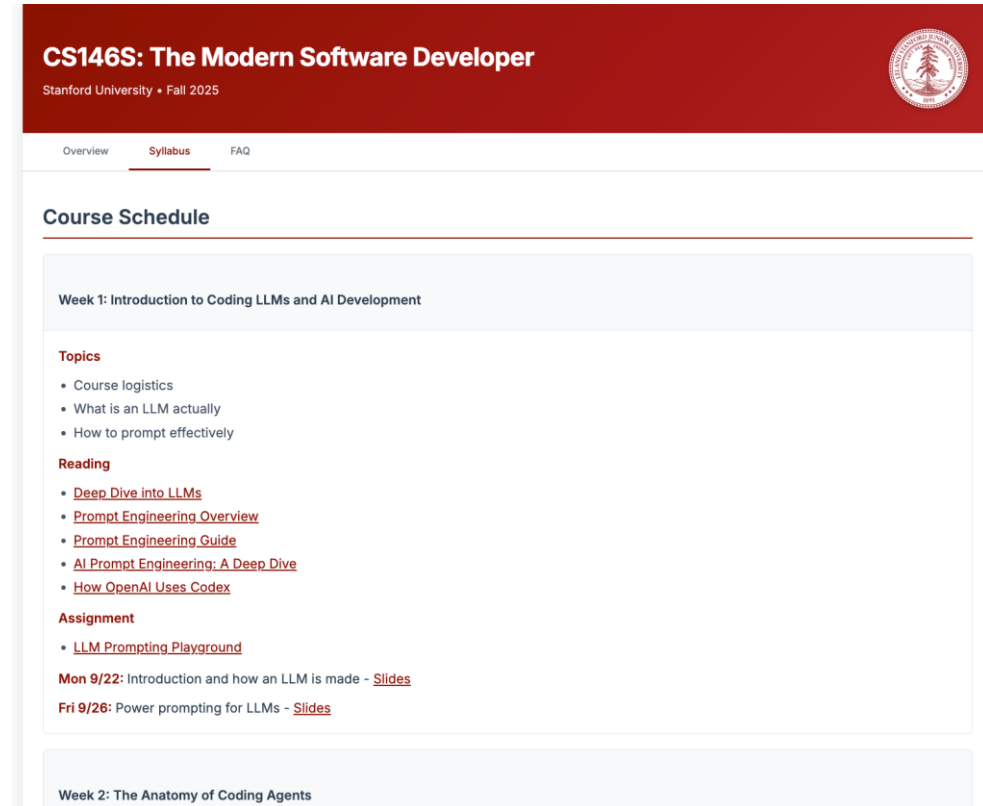
https://infs4205-7205.github.io/VibeCoding_Tutorial/

Assessment Items - Project

Project (35%): Personalised Multimodal Chatbot

Learning Resources for Vibe Coding

- 3 modular “101-level” Intro to use cursor (cred. Danny Wang)
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CS146S: The Modern Software Developer
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- [LLM Prompting Playground](#)

Mon 9/22: Introduction and how an LLM is made - [Slides](#)

Fri 9/26: Power prompting for LLMs - [Slides](#)

Week 2: The Anatomy of Coding Agents

Assessment Items III - Project

Project (35%):

Technical Report and Code (20') due end of **Week 10**

- Separately submitted via Blackboard
- The assignment brief, including detailed criteria and the marking rubric, will be released in Week 4 on Blackboard.

Code Interview (15') scheduled during **Week 13** (tutorial sessions)

- In-person, bring your student ID card
- Tutors may ask detailed questions regarding your implementation (~8 min pp)

If a student receives less than 5 marks in the code interview, the mark for report will be capped at 5 out of 20, regardless of the quality of the submitted report and code.

Assessment Items – Final Exam

Final Exam (30%) cut from 55% (2024) to 30%

- Every **tutorial** will deeply discuss a set of MLE interview questions
- **Closed-book**, centrally scheduled in exam week
- Short answer questions
 - Not tight to calculation / theory
 - More like a “Mock Interview” for your future tech interview preparation
- PG Student will have an additional question
- Subject to the *“double-pass”* hurdle
 - Students must receive a passing grade on the final exam to pass the course



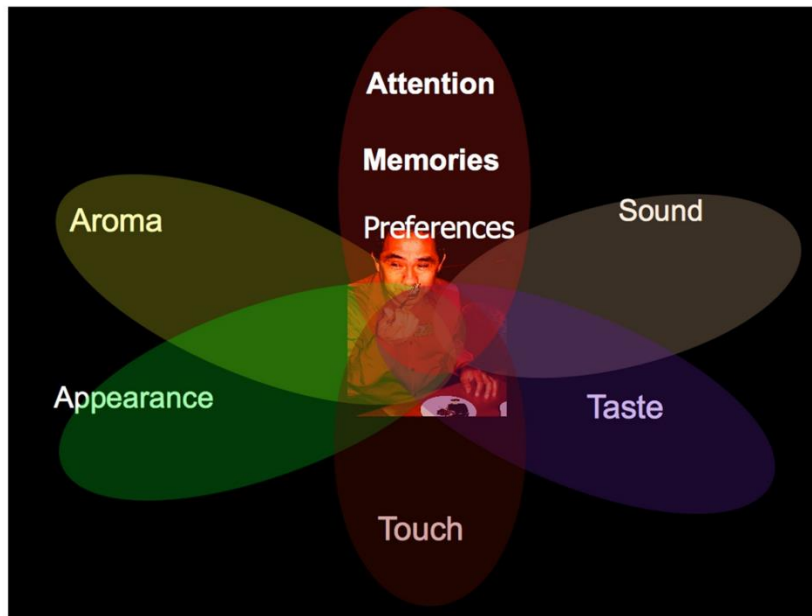
Assessment Summary

Reflective Inquiry	25 marks
In-Semester Quiz	10 marks
Project	
- Project report	20 marks
- Code interview	15 marks
Final Exam (double-pass)	30 marks
Total	100 marks

Advanced Techniques for High-dimensional Data

From the Curse of Dimensionality to Multimodal Intelligence

- The Nature of Data



Sensory Modalities

Language

- **Lexicon**
 - Words
- **Syntax**
 - Part-of-speech
 - Dependencies
- **Pragmatics**
 - Discourse acts

Acoustic

- **Prosody**
 - Intonation
 - Voice quality
- **Vocal expressions**
 - Laughter, moans

Visual

- **Gestures**
 - Head gestures
 - Eye gestures
 - Arm gestures
- **Body language**
 - Body posture
 - Proxemics
- **Eye contact**
 - Head gaze
 - Eye gaze
- **Facial expressions**
 - FACS action units
 - Smile, frowning

Touch

- **Haptics**
- **Motion**

Physiological

- **Skin conductance**
- **Electrocardiogram**

Mobile

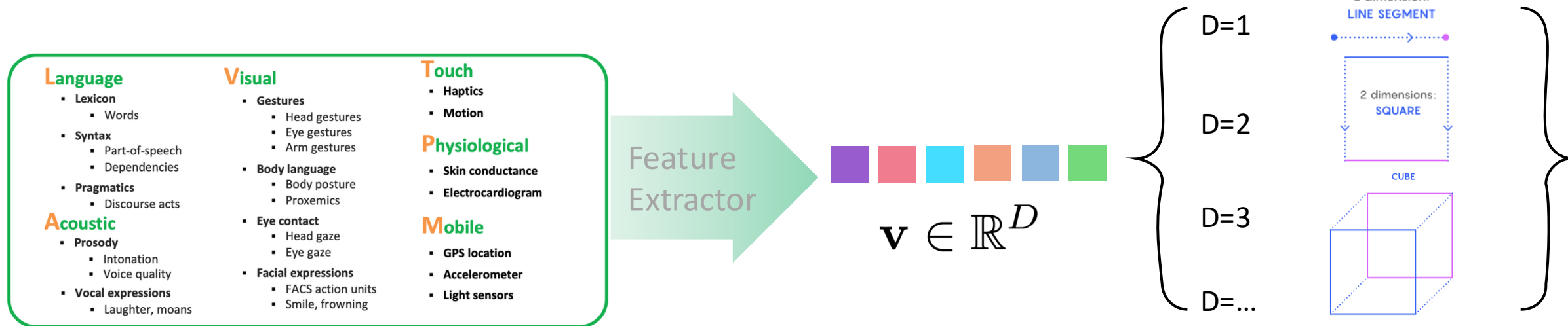
- **GPS location**
- **Accelerometer**
- **Light sensors**

Human Perception -> Biological Intelligence

Advanced Techniques for High-dimensional Data

From the Curse of Dimensionality to Multimodal Intelligence

- What is dimensionality?

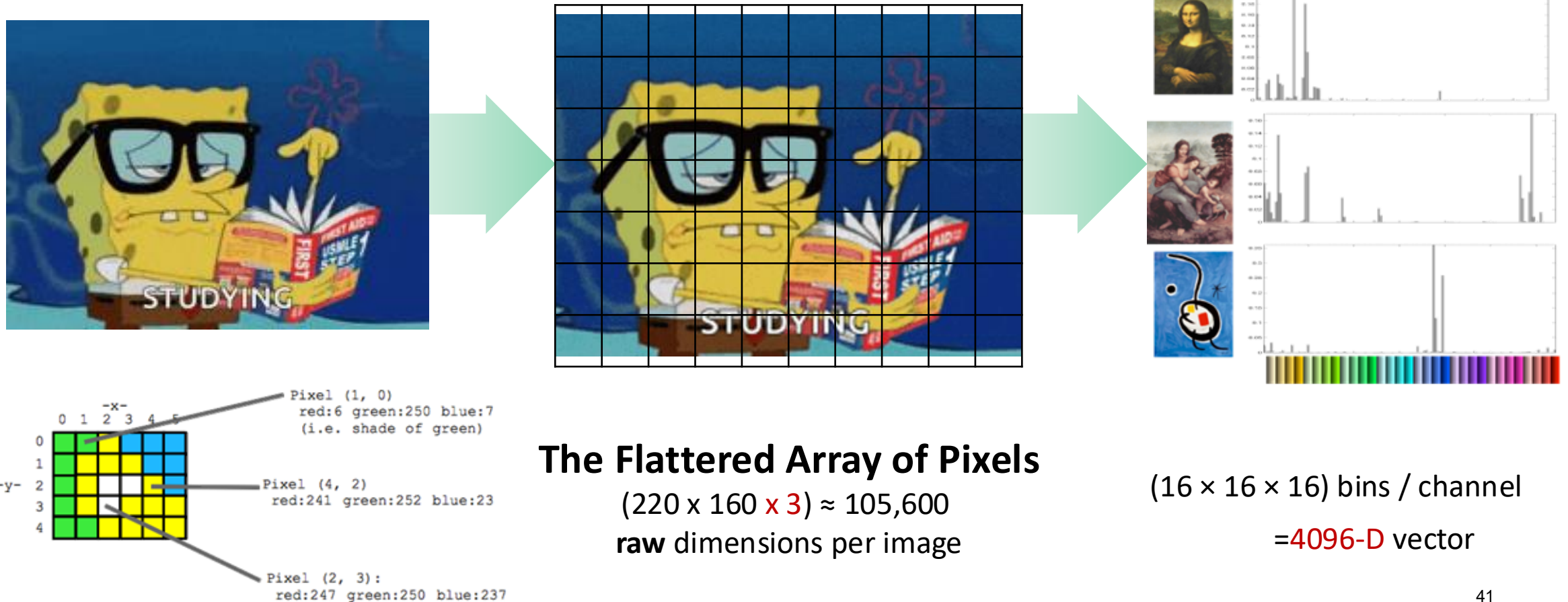


#D: the number of coordinates needed to make up a vector (**embedding**) $\mathbf{v}$

Advanced Techniques for High-dimensional Data

From the Curse of Dimensionality to Multimodal Intelligence

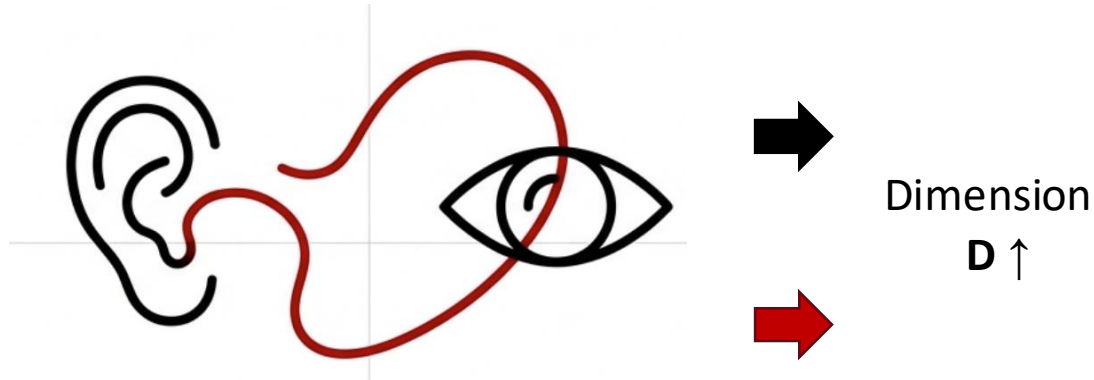
- Unimodal Embeddings are high-dimensional, discrete, fragmented



Advanced Techniques for **High-dimensional** Data

From the Curse of Dimensionality to Multimodal Intelligence

- Modalities are *coupled*, not independent



The “McGurk Effect”

demonstrates that hearing and vision are coupled, not additive.

Information from two modalities is transmitted into something new but requires high-dimensional representations

(Youtube) Audio Illusion That Will Fool Your Brain - McGurk Effect - Museum of Science



Advanced Techniques for **High-dimensional** Data

From the Curse of Dimensionality to Multimodal Intelligence

Curse of Dimensionality

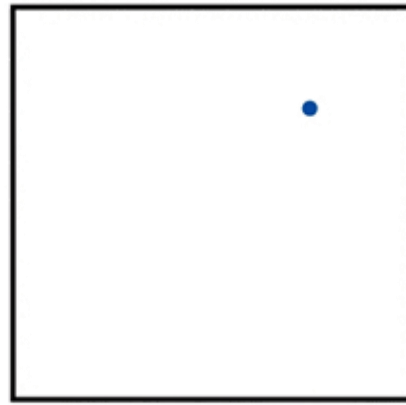
- Coined by Bellman to describe the problem **as dimensionality increases**, the **volume** of space expands exponentially, creating a **sparsity crisis**

1D: The Line



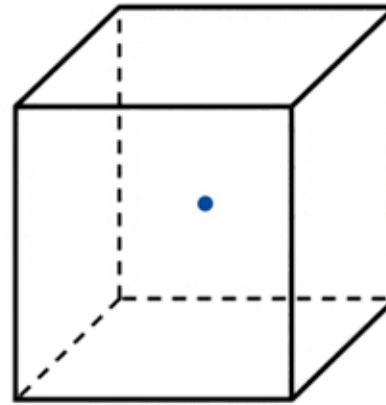
If you drop a penny on a 100-yard line, it takes minutes to find it by walking.

2D: The Field



If you drop a penny on a 100x100 yard square, it is like searching two football fields. It could take days.

3D: The Cube



If you drop a penny in a 100x100x100 yard cube, it is like searching a 30-story building the size of a football stadium.



Richard Bellman

Advanced Techniques for High-dimensional Data

From the Curse of Dimensionality to Multimodal Intelligence

1. High-dimensional Spaces are Empty

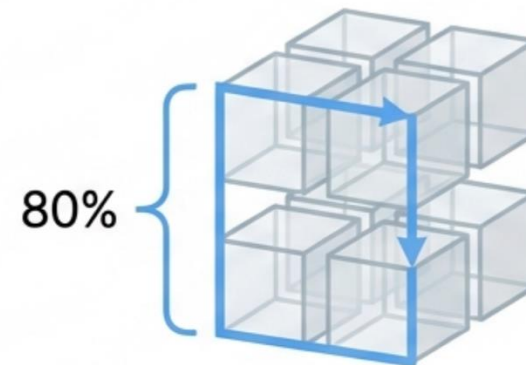
1 Dimension



To capture 10% volume, you need 10% of the edge.

$$0.1^1 = 0.1$$

10 Dimensions



To capture 10% volume, you need 80% of the edge.

$$0.8^{10} \approx 0.1$$

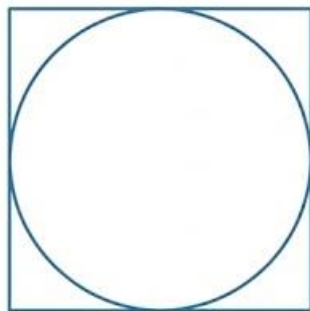
In 10 dimensions, to capture a small fraction of the data, you must cover **almost the entire range of the variable**. Points are not clustered; they are isolated on the edges.

Advanced Techniques for High-dimensional Data

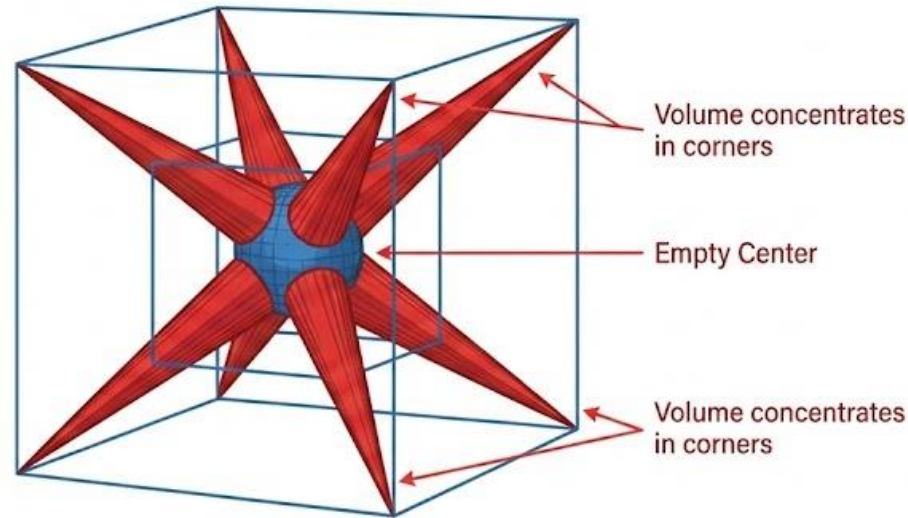
From the Curse of Dimensionality to Multimodal Intelligence

2. High-dimensional Geometry is Counter-intuitive

The Spiky Sphere Paradox



Our 3D intuition suggests a **sphere** occupies most of a cube.



$$\text{As } d \rightarrow \infty, \frac{\text{Volume(Hypersphere)}}{\text{Volume(Hypercube)}} \rightarrow 0$$

In high dimensions, the sphere is insignificant. Almost all the volume of a high-dimensional space **exists in the "corners"** -> the extreme values of the variables.

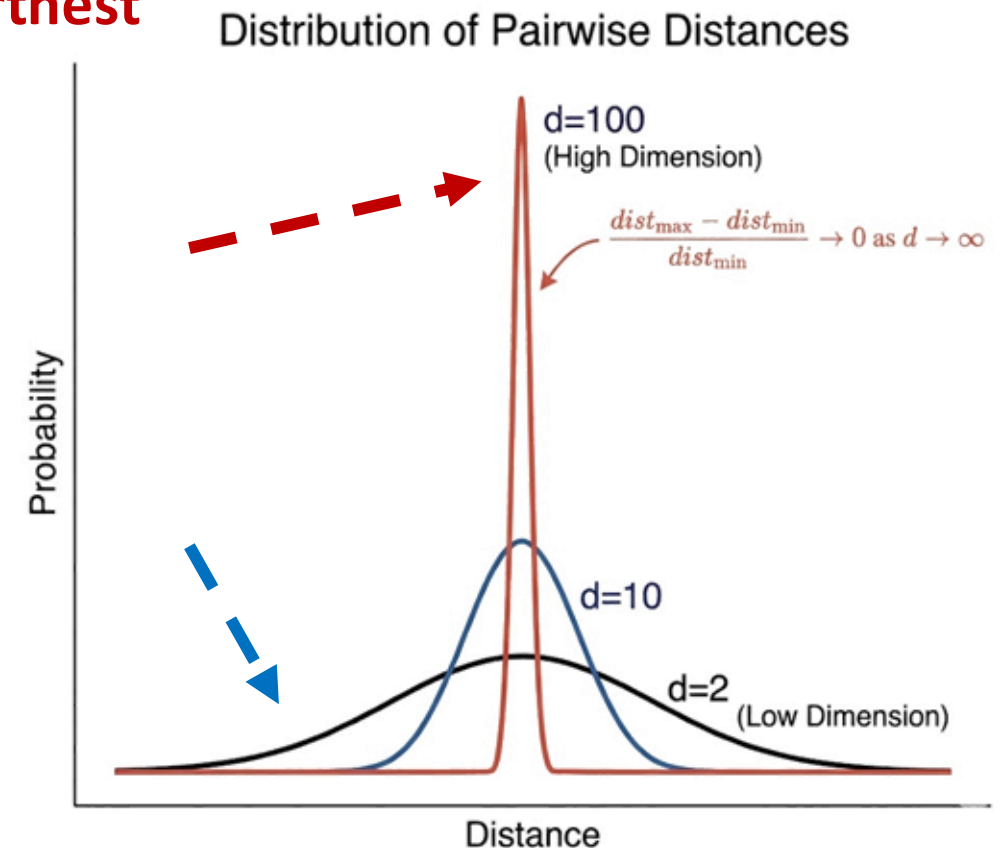
Advanced Techniques for High-dimensional Data

From the Curse of Dimensionality to Multimodal Intelligence

3. High-dimensional Nearest Neighbour is also Farthest

(Low-D) Distances are spread out. Neighbors are distinct

(High-D) the distance between the "nearest" neighbor and the "farthest" neighbor approaches zero. All points look equidistant. The concept of "closeness" vanishes.

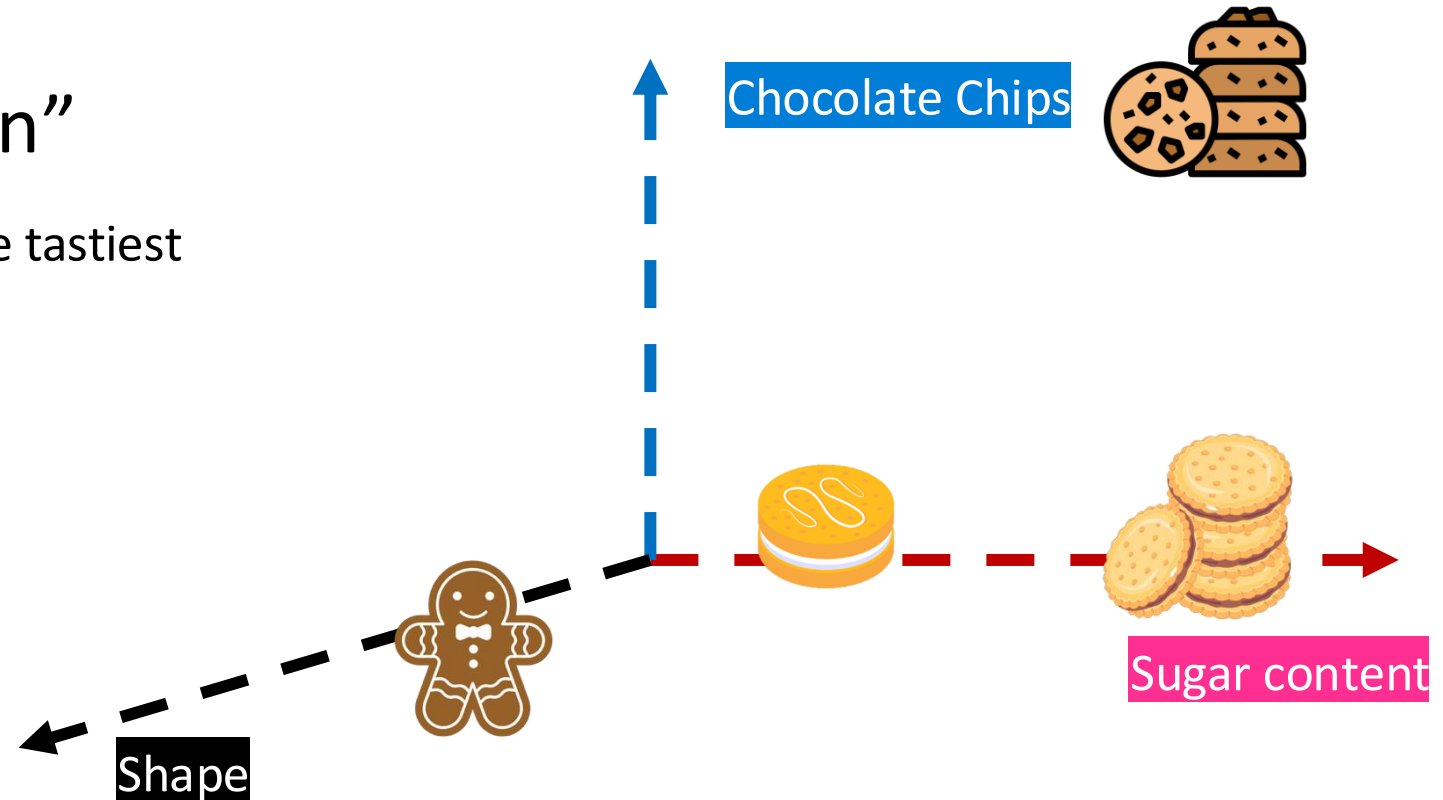


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The “Cookie Intuition”

Imaging we are trying to find the tastiest cookie...



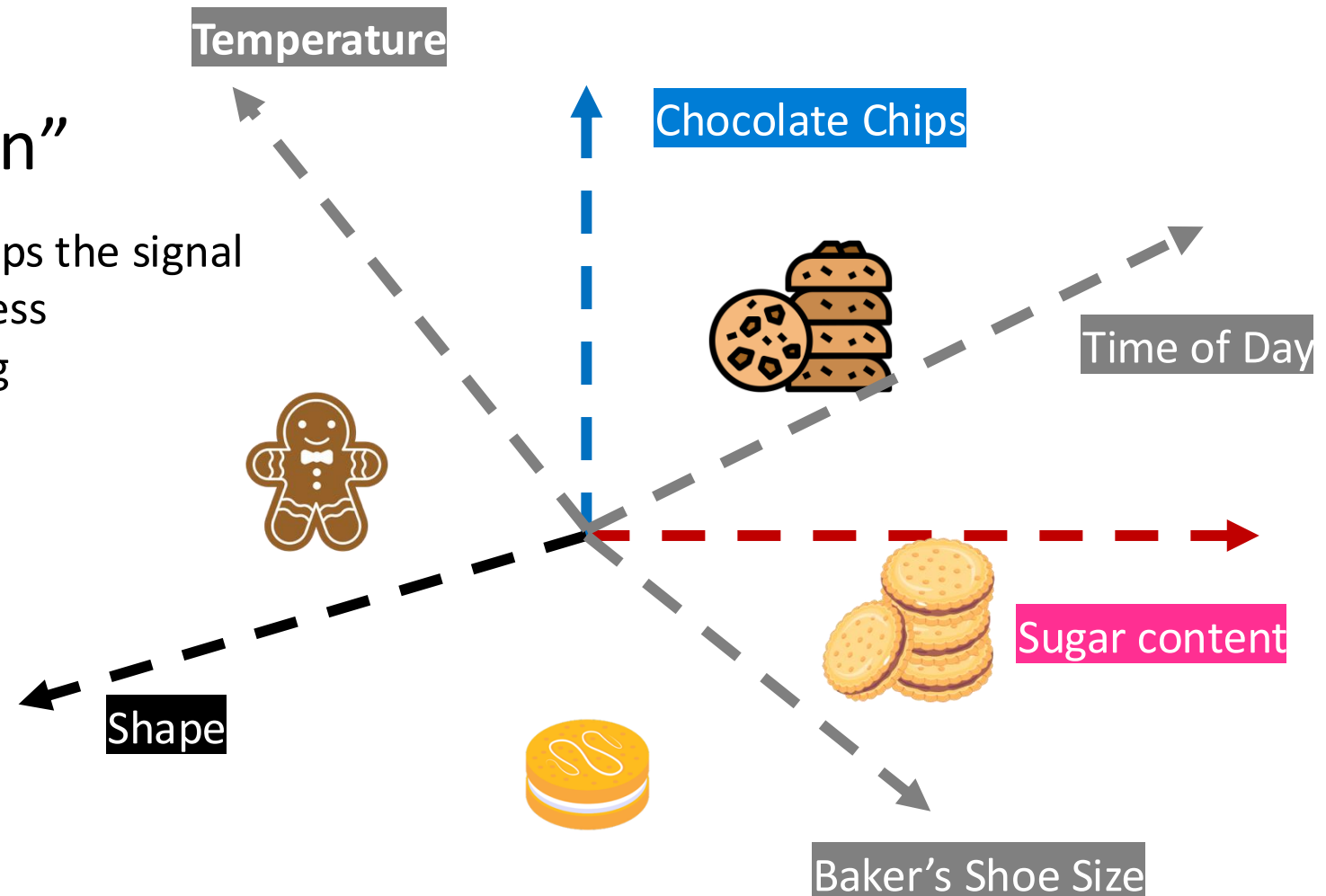
Advanced Techniques for High-dimensional Data

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The “Cookie Intuition”

In high-dimensions, **noise** swamps the signal

- Distances become meaningless
- Worse than random guessing



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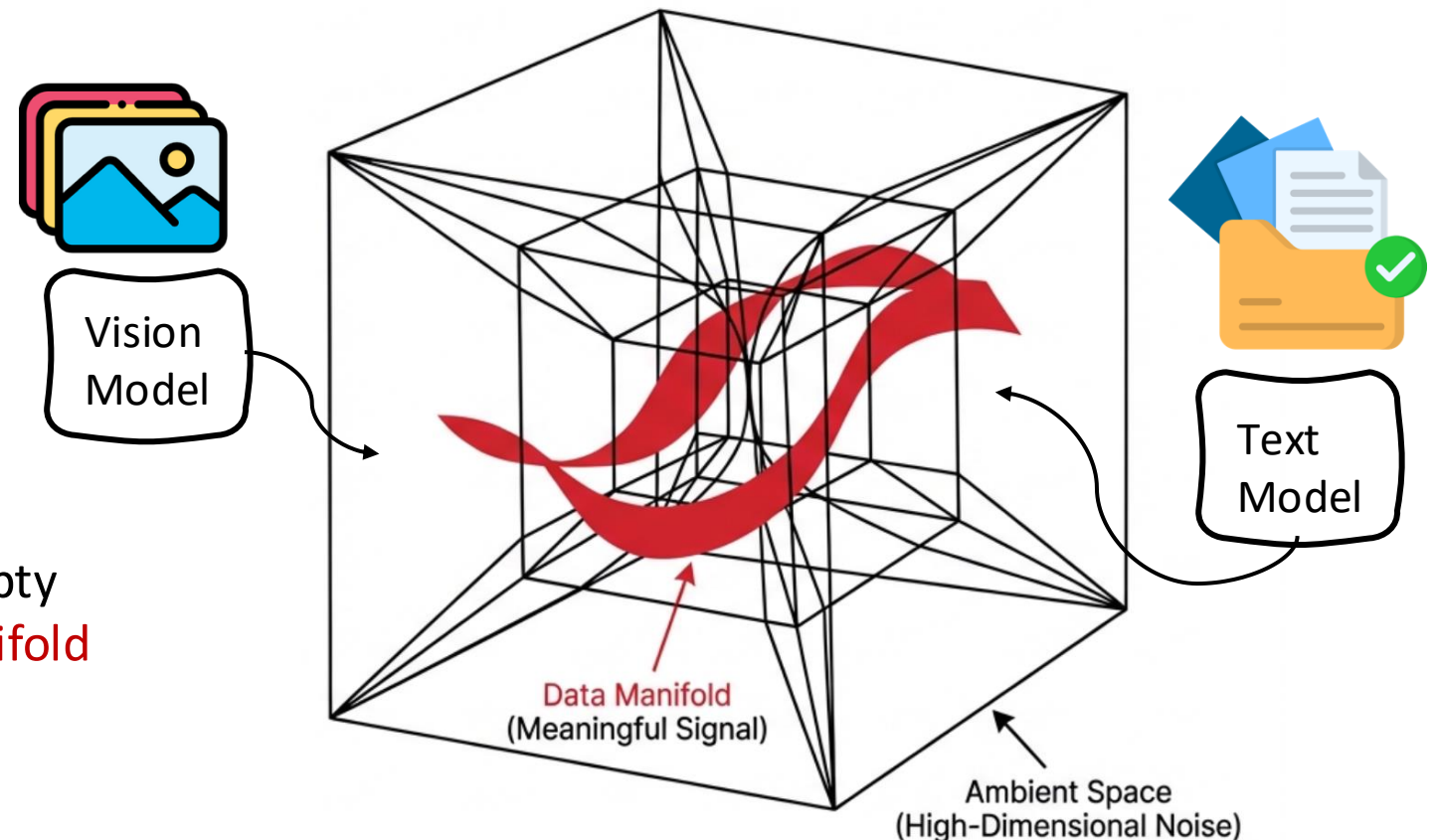
From the Curse of Dimensionality to Multimodal Intelligence

The Twist – From Curse to Blessing of Dimensionality

Why Deep Learning Works

Real-world data lies on low-dimensional manifolds embedded within high-dimensional space.

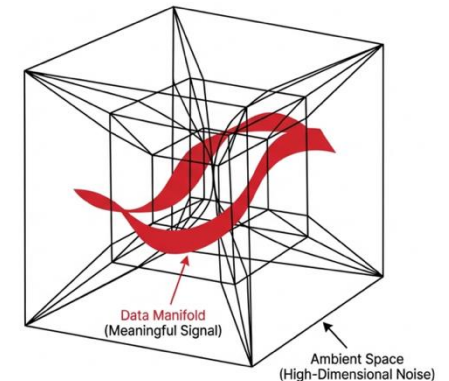
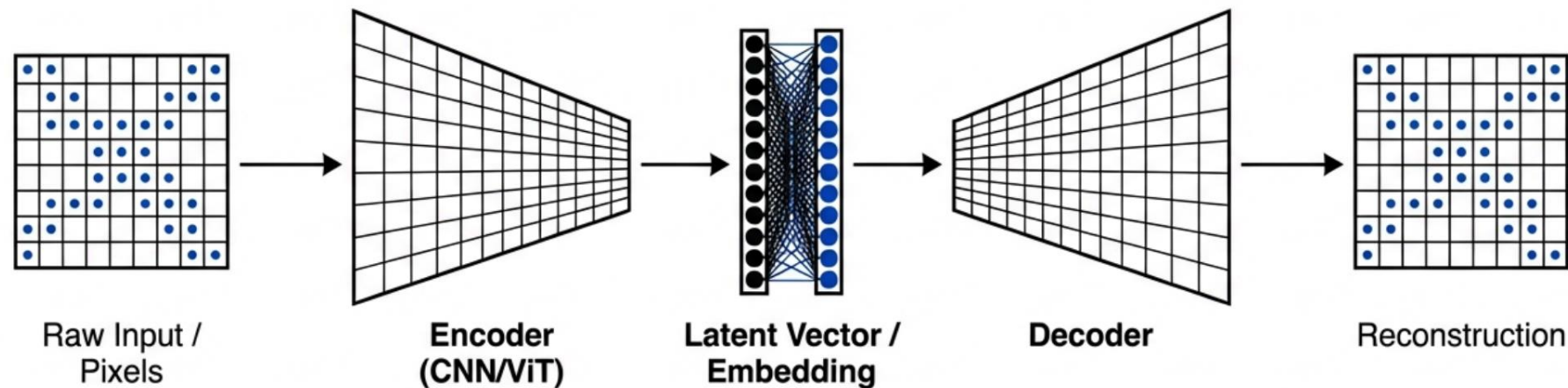
We don't need to model the entire empty space -> learn the *mapping* to the manifold



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From the Curse of Dimensionality to Multimodal Intelligence

Multimodal Representation Learning (Week 2-3)



How to learn intrinsic features?

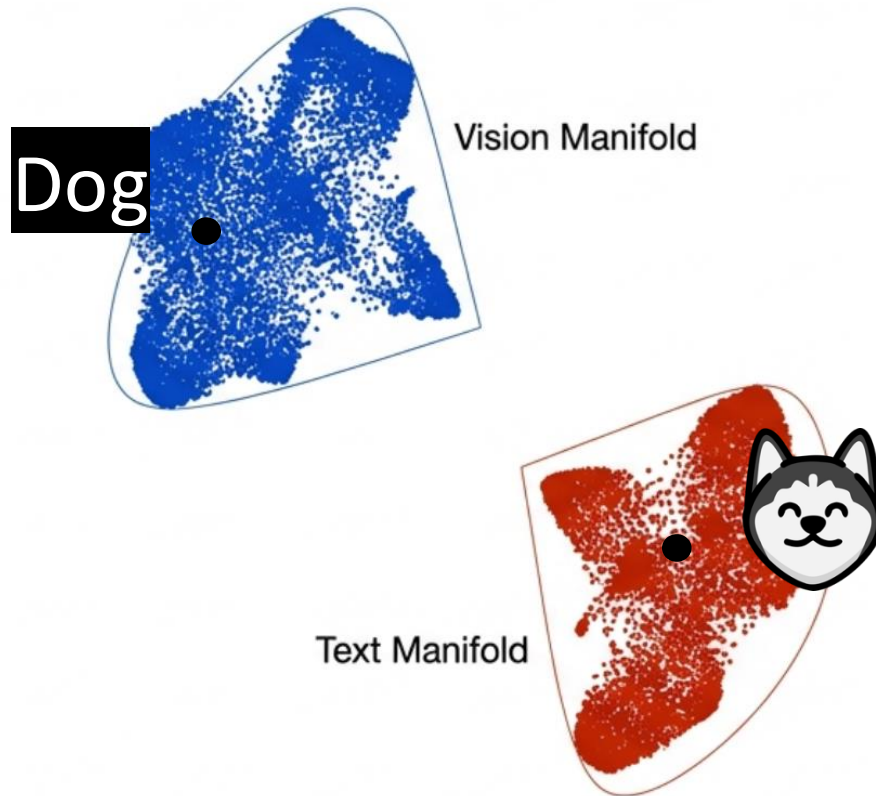
Encoder: ignore the empty space and map raw data directly to the manifold

- **Text:** transformers convert symbols to semantic vectors
- **Vision:** CNNs/ViTs map pixel grids to feature vectors

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From the Curse of Dimensionality to Multimodal Intelligence

Multimodal Alignment (Week 4)



The Challenge

Visual and text embeddings reside in different regions of the vector space. A machine cannot initially relate a “Dog” image to the word “Dog”.

-> Suboptimal cross-modal interaction without alignment

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Multimodal Alignment (Week 4) – Contrastive Learning

Mechanism:

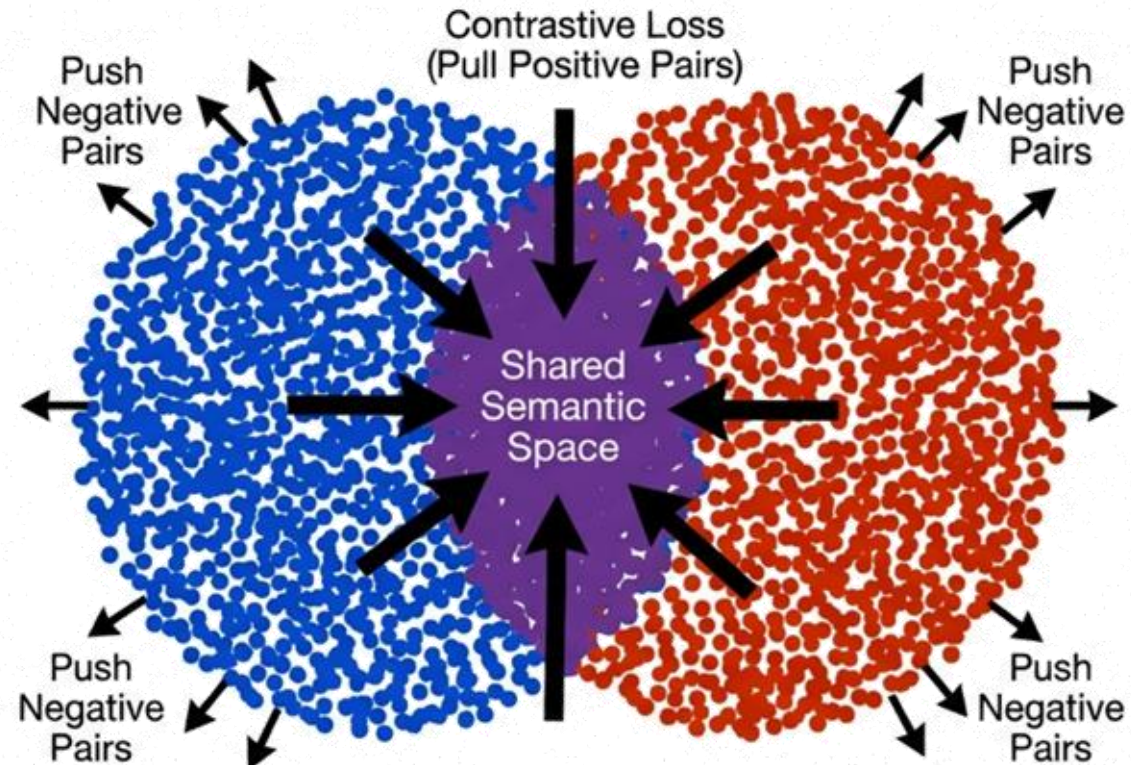
Train on massive image-text pairs.

Geometry:

Maximize cosine similarity between matched pairs; minimize it for mismatched pairs.

Result:

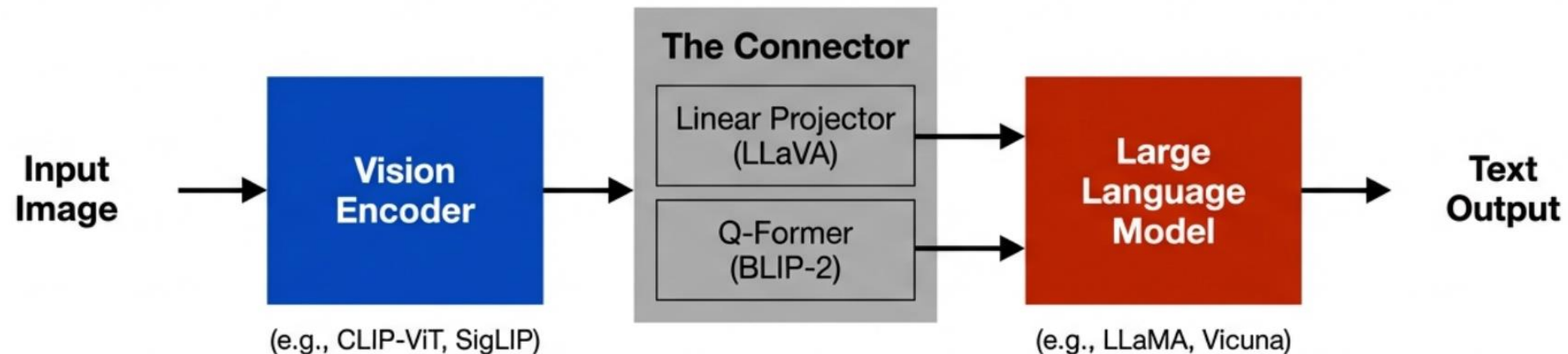
The 'Dog' image vector and 'Dog' text vector become neighbors. Geometry now equals Meaning.



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Multimodal Large Language Model (Week 5) – MLLM Architecture



The Connector's Role

Acts as a translator, projecting visual features into the language embedding space.

Q-Former

A Transformer bottleneck using learnable queries to extract visual information.

Projector

A simple MLP that aligns feature dimensions with word embeddings.

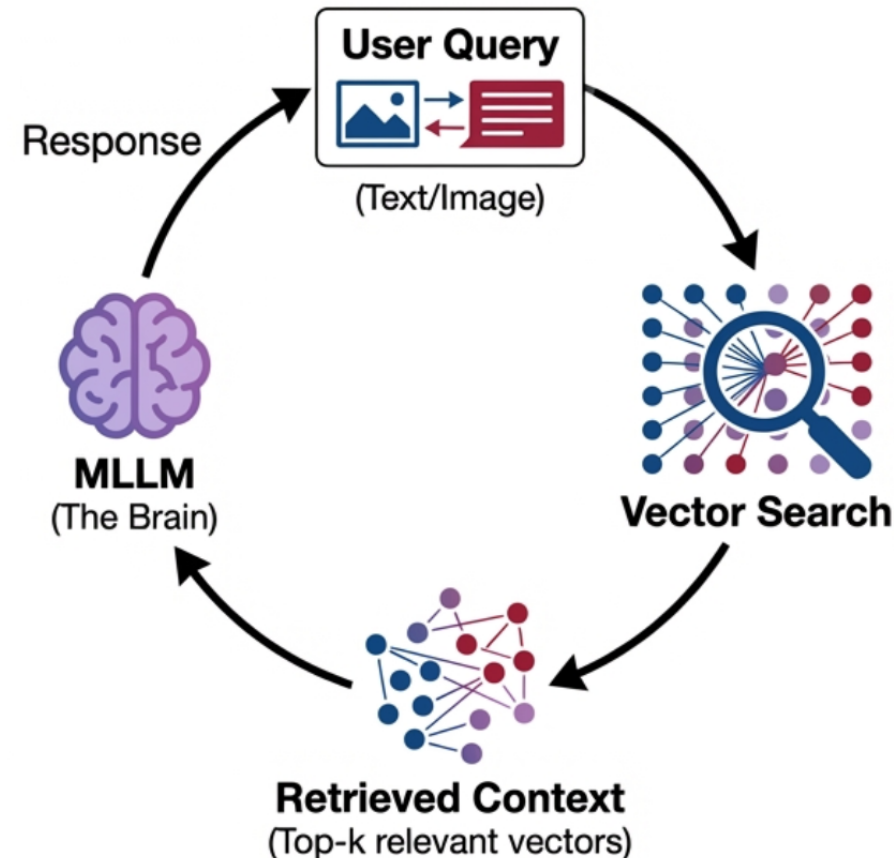
Advanced Techniques for High-dimensional Data

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High-dimensional Data Indexing (Week 6) – VectorDB

Closing the Loop

We conquered the Curse of Dimensionality, so distance metrics (Cosine Similarity) now work effectively.



Retrieval Augmented Generation (RAG)

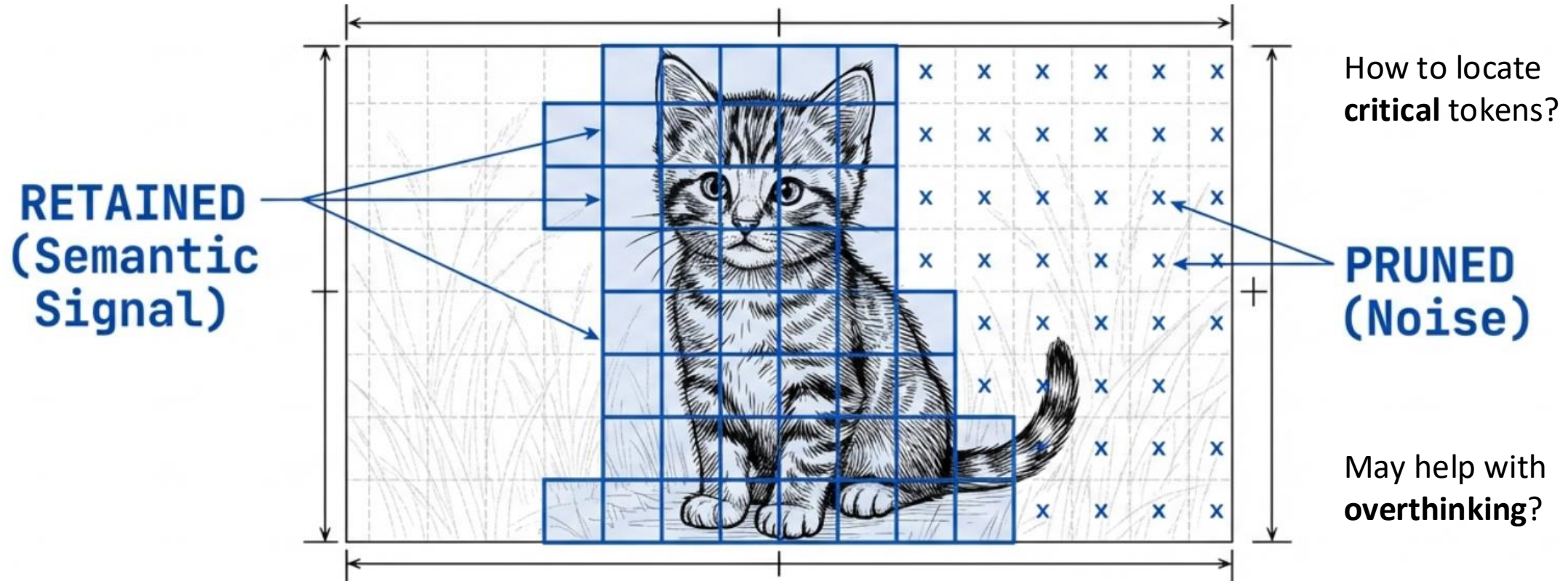
RAG Workflow

1. **Retrieve:** Query VectorDB for relevant multimodal contexts.
2. **Augment:** Inject context into the prompt.
3. **Generate:** Grounded answers with reduced hallucination.

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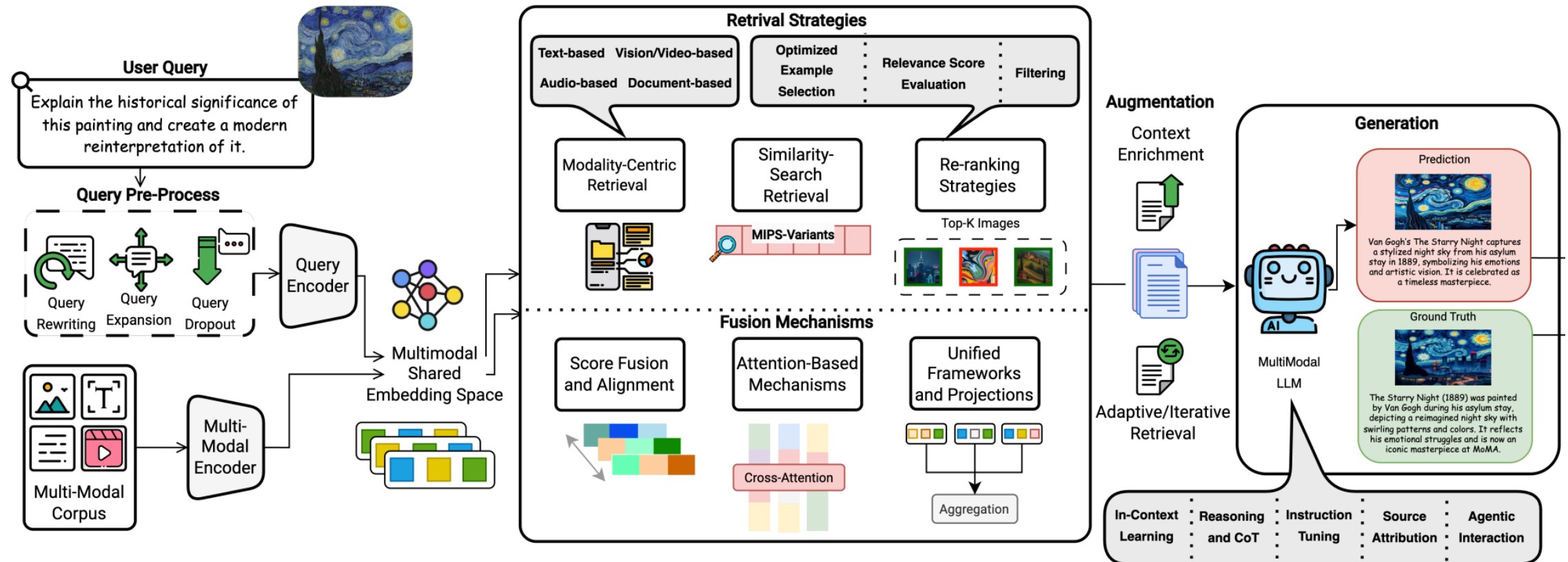
High-dimensional Data Efficiency (Week 7) – Token Reduction



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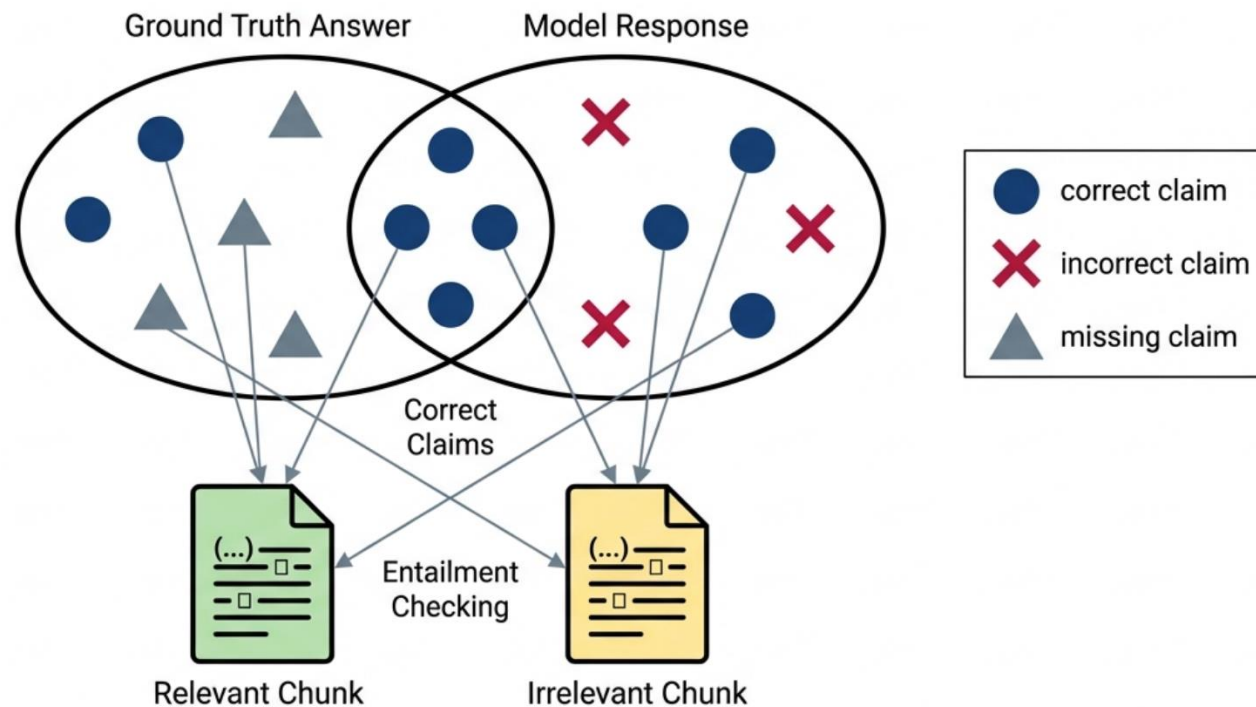
Multimodal RAG Pipeline (Week 8)



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Multimodal RAG Evaluation (Week 9)



RAGChecker Diagnostic Metrics

Retriever Diagnostics

Claim Recall: Did we retrieve all necessary facts? (% of GT claims covered).

Context Precision: Was the retrieval useful? (% of chunks containing relevant claims).

Generator Diagnostics

Context Utilization: Did the LLM use the given facts?

Noise Sensitivity: Did irrelevant chunks confuse the LLM?

Hallucination: Did the LLM invent claims not in context?

Self-Knowledge: Did the LLM rely on pre-training instead of retrieval?

Faithfulness: Ratio of supported claims vs total claims.

Advanced Techniques for **High-dimensional** Data

From the Curse of Dimensionality to Multimodal Intelligence

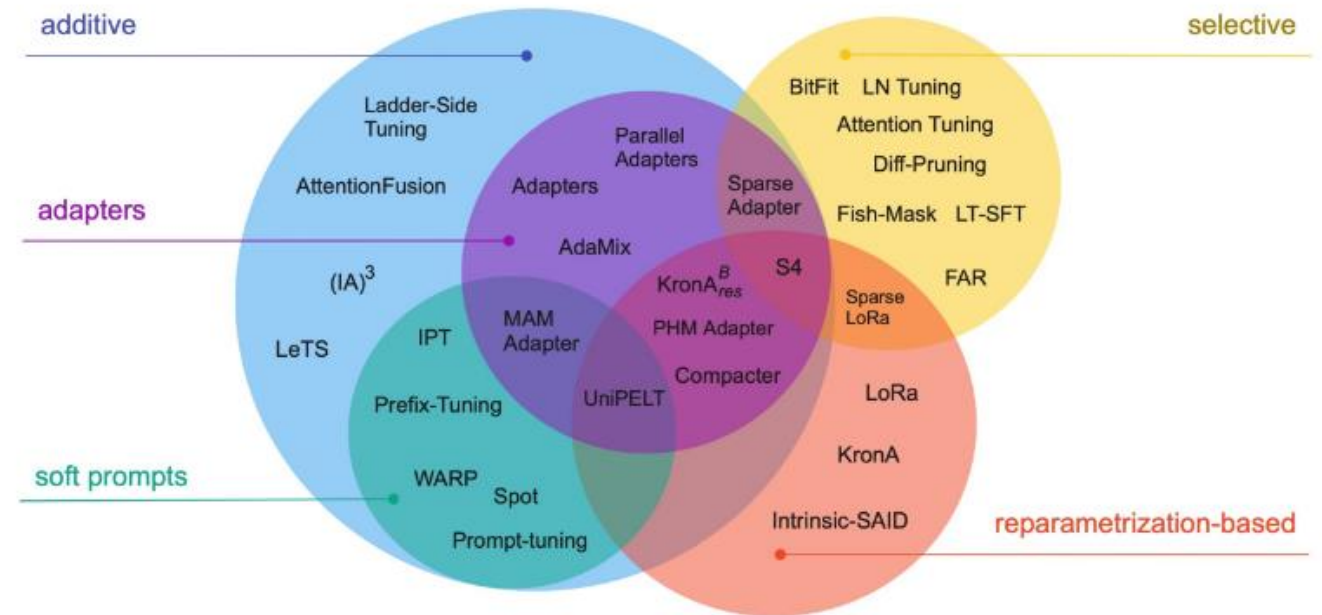
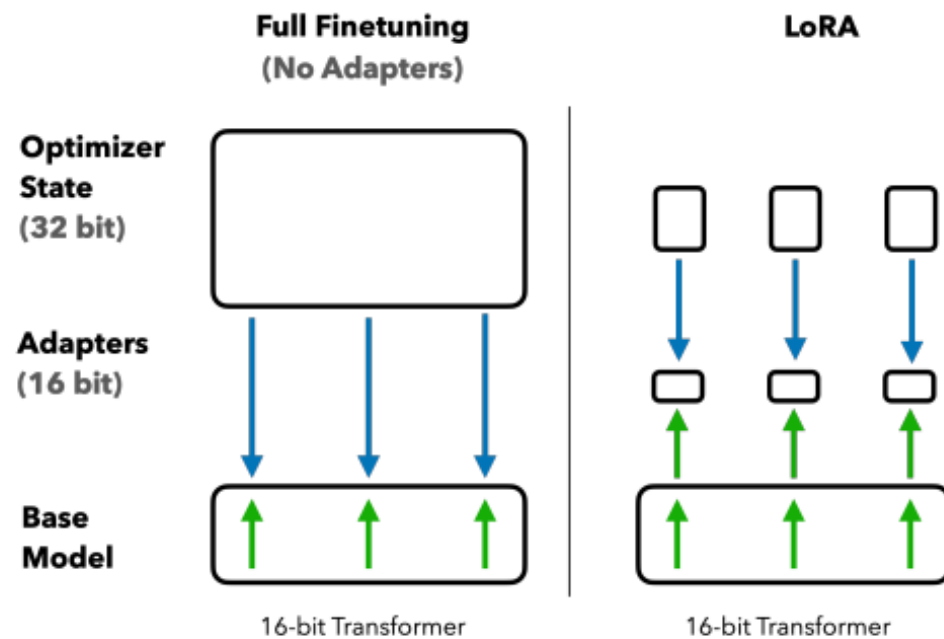
Parameter-Efficient Fine-Tuning (Week 11)

	Fine Tuning	RAG
01 Latency & Computational Cost	Requires high initial training costs but delivers fast inference.	Cheaper to train but incurs retrieval delays, leading to 30 – 50% longer response times compared to fine tuned models.
02 Scalability & Maintenance	Needs periodic retraining to stay updated.	Easier to maintain as knowledge is externally updated.
03 Knowledge Retention & Hallucination Risk	Encodes knowledge directly into the model, reducing hallucinations.	Relies on retrieval accuracy; poor indexing increases hallucination risks.

Advanced Techniques for High-dimensional Data

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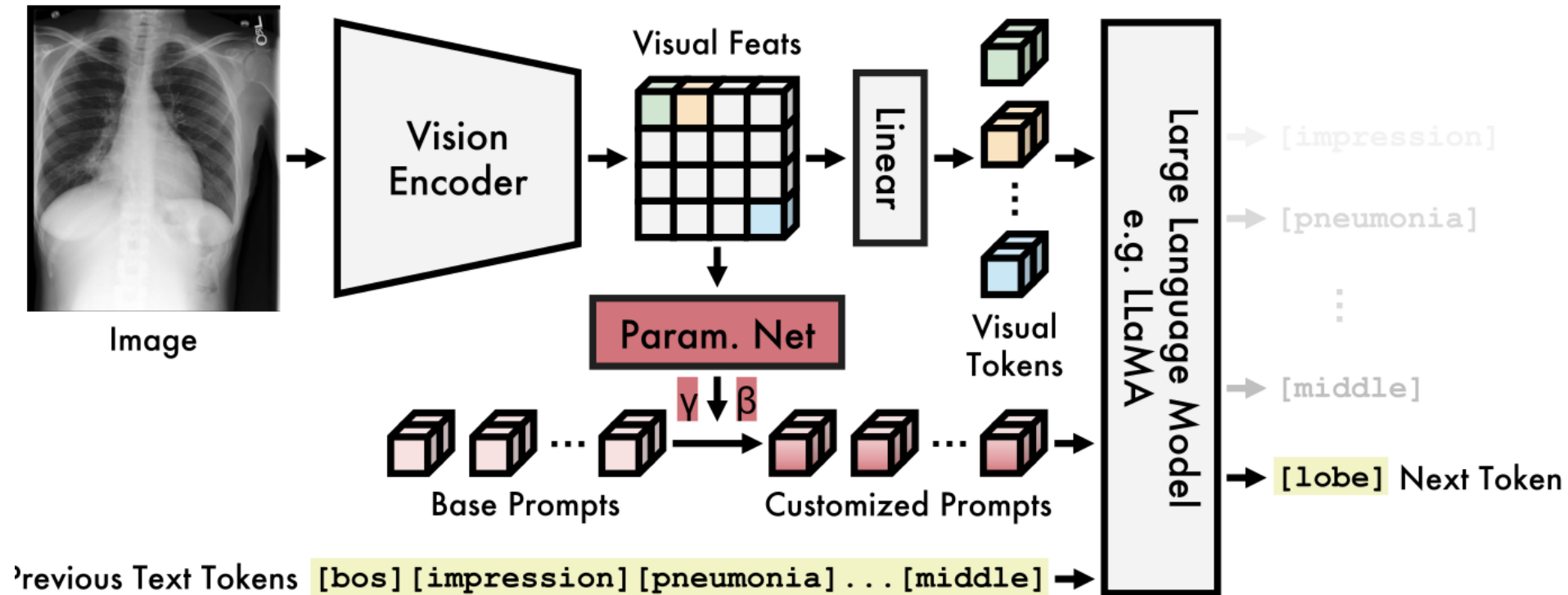
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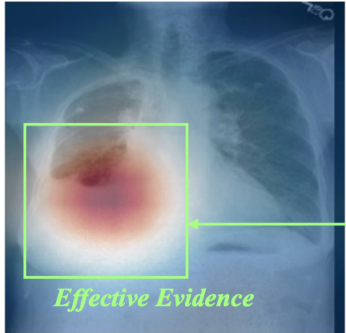
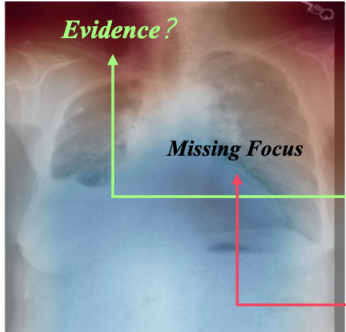
Applications (Week 12) – Medical Report Generation



Advanced Techniques for High-dimensional Data

From the Curse of Dimensionality to Multimodal Intelligence

Applications (Week 12) – Medical Report Generation

 Effective Evidence Anatomical Attention Map	Anatomies: Lungs, Mediastinum, Abdomen Pathology Entities: Atelectasis, Edema, Lung Opacity, Pleural Effusion, Pneumonia. Ground Truth: impression: mild pulmonary vascular congestion with moderate to large right pleural effusion and small left pleural effusions. right basilar opacification may reflect atelectasis and/or infection. findings: the cardiac silhouette size remains mildly enlarged. patient is status post right upper lobectomy and right upper chest wall resection with evidence of volume loss in the right lung and posttreatment changes in the right upper lung field unchanged. left hilar enlargement is unchanged with mild pulmonary vascular congestion present. moderate to large right pleural effusion and small left pleural effusion are again demonstrated not significantly changed in the interval. right basilar opacification is similar. no pneumothorax is identified. the aorta remains tortuous and calcified . <i>Positive Findings: Lung Opacity, Pleural Effusion</i>
 Evidence? Missing Focus Attention Map w/o prompts	SS-ACL: impression : interval increase in size of right pleural effusion now moderate to large in size with adjacent right basilar opacity likely reflective of compressive atelectasis though infection is not excluded. mild pulmonary vascular congestion. findings : the patient is status post right upper lobectomy with clips noted in the right hilum. heart size is difficult to assess given the presence of a moderate to large right pleural effusion increased in size compared to the previous exam. <i>Positive Findings: Lung Opacity, Pleural Effusion, BLEU-1: 0.380, F1: 0.86</i> R2GenGPT: impression : small right pleural effusion with overlying atelectasis . findings : pa and lateral views of the chest are compared to previous exam from . there is a small right pleural effusion with overlying atelectasis . the left lung is clear . cardiomeastinal silhouette is unchanged . no pneumothorax is seen . bony structures are intact . surgical clips project over the right upper quadrant . <i>Positive Findings: Atelectasis, Pleural Effusion, BLEU-1: 0.282, F1: 0.4</i> SS-ACL (w/o anatomical prompt): impression : interval increase in right pleural effusion with overlying atelectasis underlying consolidation not excluded. findings : frontal and lateral views of the chest were obtained. there has been interval increase in right-sided pleural effusion with overlying atelectasis underlying consolidation is not excluded. the left lung is clear. there is persistent blunting of the right costophrenic angle. the cardiac and mediastinal silhouettes are stable. <i>Positive Findings: Atelectasis, Pleural Effusion, BLEU-1: 0.258, F1: 0.4</i>

Advanced Techniques for **High-dimensional** Data

From the Curse of Dimensionality to Multimodal Intelligence

What you will expect from this course?

- **Hands-on practice** on multimodal RAG
- Deeper understanding on **high-dimensional** data
- Challenges and Opportunities behind
- **Mock interview questions** for AI engineer job
- **Vibe coding** on your first AI project
- Research training on **exploring the boundary of unknowns**



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OF QUEENSLAND
AUSTRALIA

CREATE CHANGE

THANKS!

Do you have any questions?

uqyluo@uq.edu.au

The University of Queensland

